Sampling, constraints, and selected-mode resolvent analysis for a long-box Chebyshev-Fourier DNS of turbulent channel flow

Prepared for the MKM long-box channel-flow workflow

July 13, 2026

Abstract

This note gives a self-contained description of a long-box MKM channel-flow simulation in `shenfun`, its statistical and dense temporal sampling records, and its selected-mode resolvent analysis. The streamwise period is increased from 2π to 4π, while the streamwise node count is doubled so that the physical grid spacing is unchanged. The discussion is organized at the level needed to reproduce the stored HDF5 objects. It covers the spin-up and stationary sampling procedure, constant-bulk-flux scaling, mean and Reynolds-stress estimators, horizontal modal covariances, a separate dense temporal continuation, and the solver-native Chebyshev-Gauss incompressibility and no-slip constraint. It then constructs the energy-orthonormal admissible velocity basis, the reduced channel resolvent, the sparse and dense temporal cross-spectral-density estimates, and the DNS projection fractions used to assess selected response modes. Production figures and numerical audits are included for both newly accessible low-streamwise-wavenumber modes and physical-wavenumber-matched comparisons with the earlier 2π box.

Keywords: channel flow; direct numerical simulation; long periodic box; Chebyshev-Fourier method; modal covariance; incompressibility constraint; resolvent analysis.

1 Purpose and scope

The objective of the workflow described here is to collect and postprocess a direct numerical simulation (DNS) of plane channel flow into a target data layout suitable for horizontally periodic, wall-normal inhomogeneous modeling. The DNS is run using the MKM channel-flow demonstration provided by `shenfun`. The postprocessed data are stored in horizontal Fourier space and retain the wall-normal dependence explicitly. For each horizontal mode, the stored quantities are compatible with a discrete incompressibility and no-slip constraint matrix.

This note focuses on the numerical method and the exact objects generated by the scripts in

`scripts/mkm_dns_target`.

The current production data set is generated by a two-stage run. A first stage develops the flow from the MKM demonstration initialization and writes no target snapshots. A second stage restarts from the accepted checkpoint and writes the velocity snapshots used for statistics and covariance estimation. The short four-snapshot run used earlier to validate the mechanics of this workflow is superseded here by the production run described in Section 5.

2 Physical and numerical setup

2.1 Channel geometry and variables

Let $x \in [0, L_x)$ denote the streamwise coordinate, let $y \in [0, L_y)$ denote the spanwise coordinate, and let $z \in [-1, 1]$ denote the wall-normal coordinate. The considered channel has

$$L_x = 4\pi, \qquad L_y = \pi.$$

The velocity is written in the target component order

$$\boldsymbol{u} = (u_x, u_y, u_z)^T,$$

where u_x is streamwise velocity, u_y is spanwise velocity, and u_z is wall-normal velocity. In these variables the incompressibility constraint is

$$\partial_x u_x + \partial_y u_y + \partial_z u_z = 0. \tag{1}$$

The no-slip condition at both walls is

$$\boldsymbol{u}(x, y, -1, t) = \mathbf{0}, \qquad \boldsymbol{u}(x, y, 1, t) = \mathbf{0}. \tag{2}$$

The solver advances the incompressible Navier-Stokes equations in a constant-flux channel configuration at friction Reynolds number

$$Re_\tau = 180.$$

In nondimensional form, a representative statement of the equation being advanced is

$$\nabla \cdot \boldsymbol{u} = 0, \tag{3}$$

$$\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla p + \frac{1}{Re_\tau} \Delta \boldsymbol{u} + f_x(t) \boldsymbol{e}_x, \tag{4}$$

with periodicity in x and y and no-slip walls in z. The MKM implementation uses the Kim-Moin-Moser channel-flow formulation in which the wall-normal velocity and wall-normal vorticity are advanced spectrally, while the remaining velocity components are reconstructed from incompressibility and the zero-horizontal-wavenumber momentum equations. This note does not require the internal solver variables; the sampled and constrained objects are written in the physical velocity component order (u_x, u_y, u_z).

2.2 Nondimensionalization used in the figures

The natural scales for the present channel-flow data are the channel half-height h and the friction velocity u_τ. In the MKM configuration used here,

$$h = \frac{1 - (-1)}{2} = 1, \qquad u_\tau = 1, \qquad \nu = \frac{u_\tau h}{Re_\tau} = \frac{1}{180}.$$

All figures below therefore report the nondimensional coordinates and statistics

$$x^* = \frac{x}{h}, \qquad y^* = \frac{y}{h}, \qquad z^* = \frac{z}{h}, \qquad t^* = \frac{t u_\tau}{h}, \tag{5}$$

and

$$u_i^* = \frac{u_i}{u_\tau}, \qquad R^*_{\alpha\beta} = \frac{R_{\alpha\beta}}{u_\tau^2}. \tag{6}$$

Since $h = u_\tau = 1$ in this run, these scalings leave the numerical values unchanged, but they identify the axes and colorbar values as friction-scaled channel-flow quantities. The constant-flux diagnostic gives the reference bulk velocity

$$\frac{U_b}{u_\tau} = \frac{Q}{2hL_xL_yu_\tau} \approx \frac{1237.94}{8\pi^2} = 15.6788, \tag{7}$$

which is useful as a run check but is not used as the primary plotting scale; the velocity and Reynolds-stress plots use the standard friction scaling in Eqs. (5)–(6).

2.3 Long-box equivalence controls

The historical demonstration stored the number 618.97 as an integrated box flux for the $2\pi \times \pi$ periodic domain. It is not itself a bulk velocity. Holding that number fixed while doubling L_x would halve U_b. The production runner therefore specifies the box-independent bulk velocity

$$U_b = 618.97/(4\pi^2) = 15.678693259774453$$

and constructs the integrated target from the actual domain volume,

$$Q_{\text{target}} = U_b\,|\Omega| = 1237.94. \tag{8}$$

The resolved bulk velocity is unchanged from the short-box run.

The initial streamwise perturbation also required a box-length guard. In the historical $L_x = 2\pi$ setup, the intended wall-unit wavenumber was written as the Fourier-domain endpoint divided by 200. Directly applying that expression at $L_x = 4\pi$ would change the physical perturbation wavelength. The long-box initialization instead fixes

$$\alpha^+ = 2\pi/200, \tag{9}$$

which reproduces the historical initialization in the original box and tiles the same physical disturbance scale across the longer streamwise period.

2.4 Chebyshev-Fourier grid

The current mesh is

$$(N_x, N_y, N_z) = (128, 32, 64),$$

where N_x is the number of streamwise points, N_y is the number of spanwise points, and N_z is the number of wall-normal quadrature points.

The default saved wall-normal mesh is the solver-native Chebyshev-Gauss quadrature mesh. For a general interval $[a, b]$, the nodes are

$$z_m = \frac{a+b}{2} + \frac{b-a}{2}\cos\left(\frac{(2m+1)\pi}{2N_z}\right), \qquad m = 0, \ldots, N_z - 1. \tag{10}$$

For the present channel $a = -1$ and $b = 1$. The endpoints $z = -1$ and $z = 1$ are not members of the Chebyshev-Gauss grid; this point matters for the construction of the boundary rows in Section 4.

The horizontal grids are equispaced:

$$x_j = j\frac{L_x}{N_x}, \qquad j = 0, \ldots, N_x - 1, \tag{11}$$

and

$$y_\ell = \ell\frac{L_y}{N_y}, \qquad \ell = 0, \ldots, N_y - 1. \tag{12}$$

The corresponding discrete Fourier wavenumbers are those returned by `numpy.fft.fftfreq`:

$$\kappa_p = 2\pi\,\text{fftfreq}(N_x, L_x/N_x)_p, \qquad p = 0, \ldots, N_x - 1, \tag{13}$$

$$\lambda_q = 2\pi\,\text{fftfreq}(N_y, L_y/N_y)_q, \qquad q = 0, \ldots, N_y - 1. \tag{14}$$

For this mesh,

$$\Delta x = \frac{4\pi}{128} = \frac{2\pi}{64}, \qquad \Delta y = \frac{\pi}{32}.$$

Thus the streamwise physical resolution is exactly the same as in the earlier box, while the fundamental wavenumber changes from one to one half:

$$\kappa_p = \frac{p}{2} \quad \text{on the positive branch}, \qquad \lambda_q = 2q \quad \text{on the positive branch}.$$

The longer box adds low-κ content without lowering the resolved streamwise maximum wavenumber.

2.5 Time integration and staged sampling

The production run used the following solver settings:

$$Re_\tau = 180, \qquad \Delta t = 5\times 10^{-4}, \qquad \texttt{conv} = 0,$$
$$\texttt{family} = \texttt{C}, \qquad \texttt{padding\_factor} = (1.5, 1.5, 1.5), \qquad \texttt{timestepper} = \texttt{IMEXRK222}.$$

The data collection is explicitly split into two stages.

First, a spin-up stage starts from the MKM demonstration initialization. It writes no velocity snapshots for the target data set. Its only persistent state is a checkpoint forced at the final time. The initial $0 \le t \le 20$ stage was extended after the late-time wall-normal energy block drift narrowly missed the five-percent scalar gate. The accepted production spin-up is

$$t_{\text{spin}} u_\tau/h = 40, \qquad n_{\text{spin}} = 80000.$$

Second, a sampling stage restarts from this checkpoint and writes velocity snapshots at a fixed step cadence. Snapshots were written every 200 time steps, so that

$$\Delta t_{\text{sample}} u_\tau/h = 200\Delta t\, u_\tau/h = 0.1.$$

The sampling stage was appended in four segments:

$$[40, 80], \qquad [80, 120], \qquad [120, 160], \qquad [160, 200].$$

The first segment created the sampled velocity file, and the later segments used append mode. The raw sampled velocity file therefore covers

$$tu_\tau/h = 40.1,\ 40.2, \ldots, 200.0,$$

with 1600 snapshots. No spin-up snapshots are present in that sampled velocity file. After convergence checks, the accepted target window was chosen as

$$\mathcal{T}_{\text{acc}} = \{60.0, 60.1, \ldots, 200.0\} \quad \text{in } tu_\tau/h, \qquad N_s = 1401. \tag{15}$$

Only these accepted snapshots enter the final mean, Reynolds-stress, modal coefficient, and covariance estimates.

3 Sampled velocity fields and target statistics

3.1 Component ordering

The MKM HDF5 output groups use the solver component labels

$$\texttt{u0} = u_z, \qquad \texttt{u1} = u_x, \qquad \texttt{u2} = u_y.$$

The postprocessor reorders these into the target component order

$$(u_x, u_y, u_z).$$

Let $\alpha, \beta \in \{x, y, z\}$ denote target component indices. For the n-th selected snapshot, define the sampled velocity values

$$u_\alpha(x_j, y_\ell, z_m, t_n), \qquad n = 0, \ldots, N_s - 1,$$

where N_s is the number of selected snapshots. Unless otherwise stated, the time origin in the estimators is reset to the first retained snapshot, so

$$t_n = n\Delta t_s.$$

The accepted production target uses $N_s = 1401$.

3.2 Mean profile

Because the channel is statistically homogeneous in x and y, the mean profile is a function only of the wall-normal coordinate. The estimator used in the target file is the average over selected snapshots and both horizontal directions:

$$\overline{u}_\alpha(z_m) = \frac{1}{N_s N_x N_y} \sum_{n=0}^{N_s-1} \sum_{j=0}^{N_x-1} \sum_{\ell=0}^{N_y-1} u_\alpha(x_j, y_\ell, z_m, t_n). \tag{16}$$

This is the dataset `mean_profile`, stored with shape

$$(N_z, 3).$$

3.3 Fluctuation field

The fluctuation field is formed by subtracting the estimated mean profile from each selected snapshot:

$$u'_\alpha(x_j, y_\ell, z_m, t_n) = u_\alpha(x_j, y_\ell, z_m, t_n) - \overline{u}_\alpha(z_m). \tag{17}$$

This definition ensures that the horizontal and snapshot average of u'_α is zero for each wall-normal level when the same selected snapshots are used in Eqs. (16) and (17).

3.4 Reynolds-stress profile

The stored Reynolds-stress profile is the second moment of the fluctuation field at each wall-normal level:

$$R_{\alpha\beta}(z_m) = \frac{1}{N_s N_x N_y} \sum_{n=0}^{N_s-1} \sum_{j=0}^{N_x-1} \sum_{\ell=0}^{N_y-1} u'_\alpha(x_j, y_\ell, z_m, t_n) u'_\beta(x_j, y_\ell, z_m, t_n). \tag{18}$$

The stored dataset is `reynolds_stress_profile`, with shape

$$(N_z, 3, 3).$$

3.5 Horizontal Fourier coefficients

For each snapshot and wall-normal level, the horizontal Fourier coefficient of the fluctuation field is defined by the normalized two-dimensional discrete Fourier transform

$$\widehat{u}_\alpha(z_m; \kappa_p, \lambda_q, t_n) = \frac{1}{N_x N_y} \sum_{j=0}^{N_x-1} \sum_{\ell=0}^{N_y-1} u'_\alpha(x_j, y_\ell, z_m, t_n) \exp\left[-\mathrm{i}(\kappa_p x_j + \lambda_q y_\ell)\right]. \tag{19}$$

This is the normalization used by

```
np.fft.fft2(u_fluct, axes=(streamwise, spanwise))/(Nx*Ny).
```

For a fixed horizontal mode (κ_p, λ_q), the postprocessor stacks the three components level by level into the modal vector

$$\widehat{\boldsymbol{q}}_{pq}(t_n) = \begin{bmatrix} \widehat{u}_x(z_0; \kappa_p, \lambda_q, t_n) \\ \widehat{u}_y(z_0; \kappa_p, \lambda_q, t_n) \\ \widehat{u}_z(z_0; \kappa_p, \lambda_q, t_n) \\ \widehat{u}_x(z_1; \kappa_p, \lambda_q, t_n) \\ \widehat{u}_y(z_1; \kappa_p, \lambda_q, t_n) \\ \widehat{u}_z(z_1; \kappa_p, \lambda_q, t_n) \\ \vdots \\ \widehat{u}_x(z_{N_z-1}; \kappa_p, \lambda_q, t_n) \\ \widehat{u}_y(z_{N_z-1}; \kappa_p, \lambda_q, t_n) \\ \widehat{u}_z(z_{N_z-1}; \kappa_p, \lambda_q, t_n) \end{bmatrix} \in \mathbb{C}^{3N_z}. \tag{20}$$

The dataset `modal/u_hat` stores these modal vectors with shape

$$(N_s, N_x, N_y, 3N_z).$$

For the accepted production target this is

$$(1401, 128, 32, 192).$$

3.6 Equal-time modal covariance

The equal-time covariance matrix for each horizontal mode is estimated from the selected snapshots by

$$B_{pq}^{(0)} = \frac{1}{N_s} \sum_{n=0}^{N_s-1} \widehat{\boldsymbol{q}}_{pq}(t_n)\widehat{\boldsymbol{q}}_{pq}(t_n)^*, \tag{21}$$

where $(\cdot)^*$ denotes conjugate transpose. Equation (21) is Hermitian and positive semidefinite in exact arithmetic. For numerical robustness, the stored matrix is symmetrized as

$$B_{pq,\text{stored}}^{(0)} = \frac{1}{2}\left(B_{pq}^{(0)} + \left(B_{pq}^{(0)}\right)^*\right). \tag{22}$$

The dataset `modal/B0_DNS` has shape

$$(N_x, N_y, 3N_z, 3N_z).$$

For the accepted production target this is

$$(128, 32, 192, 192).$$

3.7 Positive-lag modal covariance

For an integer lag $s \geq 0$, the positive-lag covariance uses the unbiased estimator

$$B_{pq}^{(s)} = \frac{1}{N_s - s} \sum_{n=0}^{N_s-s-1} \widehat{\boldsymbol{q}}_{pq}(t_{n+s})\widehat{\boldsymbol{q}}_{pq}(t_n)^*, \qquad 0 \leq s < N_s. \tag{23}$$

For $s = 0$, Eq. (23) equals Eq. (21). Therefore the stored `lag_0` provides an independent check against `B0_DNS`, up to the explicit Hermitian symmetrization in Eq. (22). The accepted production target saved `lag_0` and `lag_1`.

4 Derivation of the constraint matrix

This section derives the matrix $\widetilde{G}(\kappa, \lambda)$ saved by the constraint builder. The construction starts from the continuous divergence and no-slip constraints, discretizes them on the same vertical grid and component ordering used by the target file, and then compresses the possibly redundant constraint rows by singular-value decomposition.

4.1 Fourier-mode form of incompressibility

Start from the continuous incompressibility equation

$$\partial_x u_x + \partial_y u_y + \partial_z u_z = 0.$$

For one horizontal Fourier mode (κ, λ), write

$$u_\alpha(x, y, z) = \widehat{u}_\alpha(z; \kappa, \lambda) \exp[\mathrm{i}(\kappa x + \lambda y)]. \tag{24}$$

Substituting Eq. (24) into Eq. (1) gives

$$\partial_x u_x = \mathrm{i}\kappa \widehat{u}_x(z;\kappa,\lambda)\exp[\mathrm{i}(\kappa x+\lambda y)],$$
$$\partial_y u_y = \mathrm{i}\lambda \widehat{u}_y(z;\kappa,\lambda)\exp[\mathrm{i}(\kappa x+\lambda y)],$$

and

$$\partial_z u_z = \frac{\mathrm{d}\widehat{u}_z}{\mathrm{d}z}(z;\kappa,\lambda)\exp[\mathrm{i}(\kappa x+\lambda y)].$$

The exponential factor is nonzero for all x, y, so it can be divided out. The divergence-free condition for the mode is therefore

$$\mathrm{i}\kappa \widehat{u}_x(z;\kappa,\lambda) + \mathrm{i}\lambda \widehat{u}_y(z;\kappa,\lambda) + \frac{\mathrm{d}\widehat{u}_z}{\mathrm{d}z}(z;\kappa,\lambda) = 0. \tag{25}$$

4.2 Fourier-mode form of no-slip boundary conditions

The no-slip condition is

$$u_x(x,y,\pm 1) = u_y(x,y,\pm 1) = u_z(x,y,\pm 1) = 0.$$

Using the single-mode representation in Eq. (24), for example at $z = 1$,

$$u_\alpha(x,y,1) = \widehat{u}_\alpha(1;\kappa,\lambda)\exp[\mathrm{i}(\kappa x+\lambda y)].$$

Again the exponential factor is nonzero. Hence the boundary condition is equivalent mode by mode to

$$\widehat{u}_\alpha(-1;\kappa,\lambda) = 0, \qquad \widehat{u}_\alpha(1;\kappa,\lambda) = 0, \qquad \alpha \in \{x,y,z\}. \tag{26}$$

4.3 Nodal unknown vector and extraction matrices

For a fixed horizontal mode, define the nodal vectors

$$\widehat{\boldsymbol{u}}_x = \begin{bmatrix} \widehat{u}_x(z_0) & \cdots & \widehat{u}_x(z_{N_z-1}) \end{bmatrix}^T,$$

and analogously $\widehat{\boldsymbol{u}}_y, \widehat{\boldsymbol{u}}_z$. The target modal vector stacks these values level by level:

$$\widehat{\boldsymbol{q}} = \begin{bmatrix} \widehat{u}_x(z_0) \\ \widehat{u}_y(z_0) \\ \widehat{u}_z(z_0) \\ \widehat{u}_x(z_1) \\ \widehat{u}_y(z_1) \\ \widehat{u}_z(z_1) \\ \vdots \\ \widehat{u}_x(z_{N_z-1}) \\ \widehat{u}_y(z_{N_z-1}) \\ \widehat{u}_z(z_{N_z-1}) \end{bmatrix} \in \mathbb{C}^{3N_z}. \tag{27}$$

Define three extraction matrices

$$E_x, E_y, E_z \in \mathbb{R}^{N_z \times 3N_z}$$

by their nonzero entries

$$(E_x)_{m,3m} = 1, \qquad (E_y)_{m,3m+1} = 1, \qquad (E_z)_{m,3m+2} = 1, \qquad m = 0,\ldots,N_z-1, \tag{28}$$

where zero-based indexing is used in Eq. (28), as in the implementation. Multiplying Eq. (27) by these matrices selects the component nodal vectors:

$$E_x\widehat{\boldsymbol{q}} = \widehat{\boldsymbol{u}}_x, \qquad E_y\widehat{\boldsymbol{q}} = \widehat{\boldsymbol{u}}_y, \qquad E_z\widehat{\boldsymbol{q}} = \widehat{\boldsymbol{u}}_z. \tag{29}$$

These extraction matrices are pure bookkeeping operators. They do not depend on the grid, the derivative approximation, or the wavenumber.

4.4 Barycentric spectral derivative on Chebyshev-Gauss nodes

Let $z_0, \ldots, z_{N_z-1}$ be the Chebyshev-Gauss nodes in Eq. (10). The Lagrange cardinal polynomial associated with node z_j is denoted by $\ell_j(z)$, so that

$$\ell_j(z_i) = \delta_{ij}.$$

Any nodal polynomial interpolant of a scalar function f is

$$\mathcal{I}_{N_z} f(z) = \sum_{j=0}^{N_z-1} f(z_j)\ell_j(z). \tag{30}$$

Differentiating Eq. (30) and evaluating at z_i gives

$$\frac{\mathrm{d}}{\mathrm{d}z}\mathcal{I}_{N_z} f(z)\bigg|_{z=z_i} = \sum_{j=0}^{N_z-1} f(z_j)\ell_j'(z_i). \tag{31}$$

Therefore the nodal derivative matrix is

$$(D_z)_{ij} = \ell_j'(z_i).$$

The implementation computes these entries using barycentric weights

$$\omega_j = \left(\prod_{\substack{\ell=0\\ \ell\neq j}}^{N_z-1} (z_j - z_\ell)\right)^{-1}. \tag{32}$$

For $i \neq j$, the standard barycentric differentiation formula gives

$$(D_z)_{ij} = \frac{\omega_j}{\omega_i(z_i - z_j)}. \tag{33}$$

The diagonal entry is determined by differentiating the identity

$$\sum_{j=0}^{N_z-1} \ell_j(z) = 1.$$

Taking a derivative and evaluating at z_i yields

$$\sum_{j=0}^{N_z-1} \ell_j'(z_i) = 0.$$

Since $(D_z)_{ij} = \ell_j'(z_i)$, the diagonal entry must satisfy

$$(D_z)_{ii} + \sum_{\substack{j=0\\ j\neq i}}^{N_z-1} (D_z)_{ij} = 0.$$

Thus

$$(D_z)_{ii} = -\sum_{\substack{j=0\\ j\neq i}}^{N_z-1} (D_z)_{ij}. \tag{34}$$

Equations (33) and (34) define the saved `operators/D_wall` matrix for the solver-native Chebyshev-quadrature constraint.

4.5 Endpoint interpolation for no-slip rows

Because the Chebyshev-Gauss nodes exclude the walls, no-slip boundary values cannot be selected directly from the stored nodal vector. They must be interpolated from the quadrature nodes to $z = -1$ and $z = 1$.

Using the barycentric form of the interpolant in Eq. (30), for a point ξ that is not one of the nodes,

$$\mathcal{I}_{N_z} f(\xi) = \frac{\displaystyle\sum_{j=0}^{N_z-1} \frac{\omega_j}{\xi - z_j} f(z_j)}{\displaystyle\sum_{j=0}^{N_z-1} \frac{\omega_j}{\xi - z_j}}. \tag{35}$$

Equation (35) is linear in the nodal values $f(z_j)$. Therefore the interpolation row $b(\xi) \in \mathbb{R}^{1\times N_z}$ has entries

$$b_j(\xi) = \frac{\dfrac{\omega_j}{\xi - z_j}}{\displaystyle\sum_{\ell=0}^{N_z-1} \frac{\omega_\ell}{\xi - z_\ell}}, \qquad j = 0, \dots, N_z - 1. \tag{36}$$

The two-row boundary interpolation matrix is then

$$B_\partial = \begin{bmatrix} b(-1) \\ b(1) \end{bmatrix} \in \mathbb{R}^{2\times N_z}. \tag{37}$$

For any nodal component vector $\widehat{\boldsymbol{u}}_\alpha$,

$$B_\partial \widehat{\boldsymbol{u}}_\alpha = \begin{bmatrix} \widehat{u}_\alpha(-1) \\ \widehat{u}_\alpha(1) \end{bmatrix}$$

in the polynomial-interpolation sense.

4.6 Discrete divergence rows

Evaluate the modal divergence equation (25) at all N_z Chebyshev-Gauss nodes. The first term is the streamwise derivative, which is multiplication by $\mathrm{i}\kappa$ in Fourier space:

$$\mathrm{i}\kappa \begin{bmatrix} \widehat{u}_x(z_0) \\ \vdots \\ \widehat{u}_x(z_{N_z-1}) \end{bmatrix} = \mathrm{i}\kappa\, E_x \widehat{\boldsymbol{q}}.$$

Similarly, the spanwise derivative contributes

$$\mathrm{i}\lambda\, E_y \widehat{\boldsymbol{q}}.$$

The wall-normal derivative is approximated by the spectral differentiation matrix acting on the wall-normal velocity component:

$$D_z E_z \widehat{\boldsymbol{q}}.$$

Adding the three contributions gives

$$(\mathrm{i}\kappa E_x + \mathrm{i}\lambda E_y + D_z E_z)\, \widehat{\boldsymbol{q}} = \mathbf{0}. \tag{38}$$

Therefore the discrete divergence block is

$$G_{\mathrm{div}}(\kappa, \lambda) = \mathrm{i}\kappa E_x + \mathrm{i}\lambda E_y + D_z E_z \in \mathbb{C}^{N_z\times 3N_z}. \tag{39}$$

4.7 Discrete boundary rows

The no-slip conditions in Eq. (26) require the two endpoint values of each component to vanish. For the streamwise component,

$$B_\partial E_x \widehat{\boldsymbol{q}} = \mathbf{0} \in \mathbb{C}^2.$$

For the spanwise component,

$$B_\partial E_y \widehat{\boldsymbol{q}} = \mathbf{0} \in \mathbb{C}^2.$$

For the wall-normal component,

$$B_\partial E_z \widehat{\boldsymbol{q}} = \mathbf{0} \in \mathbb{C}^2.$$

Stacking these three two-row systems gives

$$G_\partial \widehat{\boldsymbol{q}} = \mathbf{0}, \qquad G_\partial = \begin{bmatrix} B_\partial E_x \\ B_\partial E_y \\ B_\partial E_z \end{bmatrix} \in \mathbb{C}^{6\times 3N_z}. \tag{40}$$

4.8 Raw constraint matrix

The divergence constraint and boundary constraint must both hold. Combining Eqs. (39) and (40) gives the raw constraint matrix

$$G_{\mathrm{raw}}(\kappa,\lambda) = \begin{bmatrix} G_{\mathrm{div}}(\kappa,\lambda) \\ G_\partial \end{bmatrix} = \begin{bmatrix} \mathrm{i}\kappa E_x + \mathrm{i}\lambda E_y + D_z E_z \\ B_\partial E_x \\ B_\partial E_y \\ B_\partial E_z \end{bmatrix} \in \mathbb{C}^{(N_z+6)\times 3N_z}. \tag{41}$$

Thus a modal velocity vector is discretely admissible if and only if

$$G_{\mathrm{raw}}(\kappa,\lambda)\widehat{\boldsymbol{q}} = \mathbf{0}. \tag{42}$$

For $N_z = 64$, the raw constraint matrix has 70 rows and 192 columns. The SVD rank audit for the validation setup gives

$$\operatorname{rank} G_{\mathrm{raw}}(0,0) = 68,$$

and

$$\operatorname{rank} G_{\mathrm{raw}}(\kappa,\lambda) = 70 \quad \text{for the remaining 4095 horizontal modes.}$$

4.9 SVD compression and the saved $\widetilde{G}$

Some rows of G_{raw} may be linearly dependent, especially at the zero-horizontal-wavenumber mode. Moreover, later algorithms often benefit from a constraint matrix with orthonormal rows. The saved compressed matrix $\widetilde{G}$ is constructed from the singular-value decomposition of G_{raw}.

Let

$$G_{\mathrm{raw}} = U\Sigma V^* \tag{43}$$

be a full singular-value decomposition. Let the numerical rank be r, as determined by the relative tolerance

$$\sigma_j > 10^{-12}\sigma_0.$$

Partition the right singular vectors as

$$V = \begin{bmatrix} V_r & V_0 \end{bmatrix},$$

where $V_r \in \mathbb{C}^{3N_z\times r}$ corresponds to the retained singular values and $V_0 \in \mathbb{C}^{3N_z\times(3N_z-r)}$ spans the numerical nullspace. The rank-r part of Eq. (43) is

$$G_{\mathrm{raw}} = U_r \Sigma_r V_r^*. \tag{44}$$

Here U_r has orthonormal columns and Σ_r is diagonal with strictly positive retained singular values. Therefore $U_r\Sigma_r$ has full column rank. For any vector $\widehat{\boldsymbol{q}}$,

$$G_{\mathrm{raw}}\widehat{\boldsymbol{q}} = \mathbf{0} \quad \Longleftrightarrow \quad U_r\Sigma_r V_r^*\widehat{\boldsymbol{q}} = \mathbf{0}.$$

Because $U_r\Sigma_r$ has full column rank, the only way the product above can vanish is

$$V_r^*\widehat{\boldsymbol{q}} = \mathbf{0}.$$

Conversely, if $V_r^*\widehat{\boldsymbol{q}} = \mathbf{0}$, then $G_{\mathrm{raw}}\widehat{\boldsymbol{q}} = \mathbf{0}$. Thus

$$G_{\mathrm{raw}}\widehat{\boldsymbol{q}} = \mathbf{0} \quad \Longleftrightarrow \quad V_r^*\widehat{\boldsymbol{q}} = \mathbf{0}. \tag{45}$$

The saved row-orthonormal constraint matrix is therefore

$$\widetilde{G}(\kappa, \lambda) = V_r^* \in \mathbb{C}^{r\times 3N_z}. \tag{46}$$

Since V is unitary, the rows of V_r^* are orthonormal:

$$\widetilde{G}\widetilde{G}^* = V_r^*V_r = I_r. \tag{47}$$

The nullspace matrix saved for representative modes is

$$N_{\mathrm{mat}} = V_0. \tag{48}$$

Every vector of the form

$$\widehat{\boldsymbol{q}} = N_{\mathrm{mat}}\boldsymbol{a}$$

satisfies the constraint, because

$$\widetilde{G}\widehat{\boldsymbol{q}} = V_r^*V_0\boldsymbol{a} = \mathbf{0}.$$

This also explains the stored nullity

$$\mathrm{nullity} = 3N_z - r.$$

4.10 Uniform finite-difference alternative

The default for the present DNS target is the Chebyshev-quadrature spectral constraint derived above. The scripts also support a uniform-grid finite difference option for intentionally uniform saved data. In that case, the component extraction matrices E_x, E_y, E_z, the wavenumbers (κ, λ), and the algebraic forms (39)–(41) do not change. Only the wall-normal grid, the derivative matrix, and the boundary matrix change.

For a uniform grid,

$$z_m = -1 + m\frac{2}{N_z - 1}, \qquad m = 0, \dots, N_z - 1.$$

The boundary matrix is a selector:

$$B_\partial = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

The finite-difference derivative matrix is constructed row by row from local polynomial exactness. If row i uses stencil nodes $\{z_{j_0}, \dots, z_{j_{w-1}}\}$, then the weights c_s for the first derivative at z_i satisfy

$$\sum_{s=0}^{w-1} c_s(z_{j_s} - z_i)^p = \begin{cases} 1, & p = 1, \\ 0, & p = 0, 2, \dots, w-1. \end{cases} \tag{49}$$

Equation (49) is exactly the linear system solved by the finite-difference helper. For the earlier comparison, $w = 7$. This option is not the default for the present Chebyshev-quadrature DNS target.

Table 1: Field-statistical block-convergence checks for candidate sampling windows. The tolerance is 0.05 for both the mean profile and the Reynolds-stress profile.

Window	Snapshots	tu_τ/h range	Mean rel.	Reynolds rel.	Accepted
full record	1600	40.1–200	5.17×10^{-3}	5.07×10^{-2}	no
retained	1401	60–200	4.03×10^{-3}	2.93×10^{-2}	yes
later start	1201	80–200	3.94×10^{-3}	3.45×10^{-2}	yes

5 Production two-stage results

The production output directory on the Linux server is

`/media/jay/data1/shenfun_dns_runs/production_mkm_Lx4pi_64_128_32_20260713`.

The accepted target file is

`MKM_production_Lx4pi_64_128_32_target_t60_t200.h5`.

The raw velocity file `MKM_production_Lx4pi_64_128_32_U.h5` contains the full appended sampling record, while the target file contains only the accepted subset identified by the convergence study below.

5.1 Run summary and sampling-window selection

Two-stage pilots with 16 and 32 MPI ranks produced velocity fields agreeing to approximately 8.3×10^{-14} in maximum absolute difference. The 32-rank layout was retained because it reduced the measured solve time. A 64-rank run was not attempted because the unchanged 32-point spanwise direction retains the earlier Fourier-decomposition restriction. The 32-rank production schedule is

$$tu_\tau/h = 0 \to 20 \to 40 \quad \text{(spin-up)},$$

$$40 \to 80, \qquad 80 \to 120, \qquad 120 \to 160, \qquad 160 \to 200 \quad \text{(sampling)}.$$

Table 1 reports the block-convergence decision used to select the accepted samples. The full sampled interval missed the 5 percent Reynolds-stress block criterion by approximately 6.6×10^{-4}. Both trimmed candidates passed. The final target uses the earliest passing window, $60 \le tu_\tau/h \le 200$, to retain the larger sample count.

Figure 1 shows the scalar sampling-stage diagnostics as relative deviations from their sampling means. The dashed line marks the accepted start time $tu_\tau/h = 60$. The bulk flux is steady to about 10^{-7} relative variation on this scale. The 12-block scalar parser also passes: its maximum first-to-last monitored change is 2.31×10^{-2}, its maximum successive change is 4.55×10^{-2}, and the maximum divergence norm is 3.43×10^{-14}. The final data set is therefore supported by both the scalar diagnostic and the field mean/Reynolds-stress criterion in Table 1.

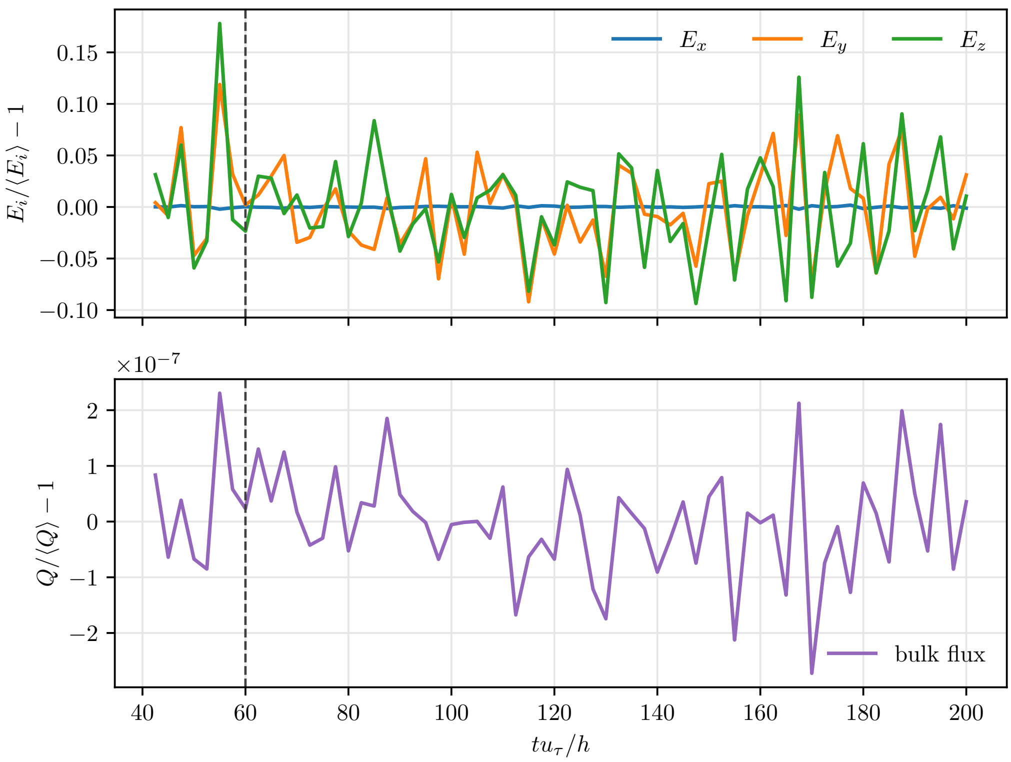


Figure 1: Sampling-stage scalar diagnostics. Upper panel: kinetic-energy components E_x, E_y, and E_z, plotted as relative deviations from their own sampling means. Lower panel: bulk flux Q, plotted as a relative deviation from its sampling mean. The abscissa is tu_τ/h, and the dashed line marks the retained-window start $tu_\tau/h = 60$.

5.2 Instantaneous and mean velocity contours

Figures 2 and 3 separate the instantaneous and mean streamwise-velocity contours. This layout preserves the equal physical aspect ratio of each slice and keeps each colorbar aligned with the height of its corresponding contour. The instantaneous $(x/h, z/h)$ slice fixes a spanwise index and displays u_x/u_τ, while the instantaneous $(y/h, z/h)$ slice fixes a streamwise index and displays the same nondimensional velocity. The mean contours are horizontally homogeneous by construction, since the estimator in Eq. (16) averages over both periodic directions.

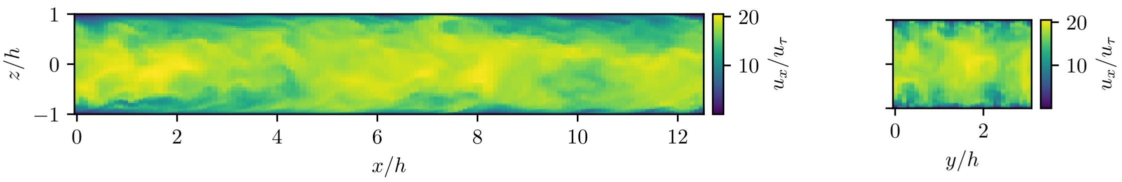


Figure 2: Instantaneous streamwise velocity u_x/u_τ at $tu_\tau/h = 200$. Left panel: streamwise-normal $(x/h, z/h)$ slice at $y/h = 0$. Right panel: spanwise-normal $(y/h, z/h)$ slice at $x/h = 0$.

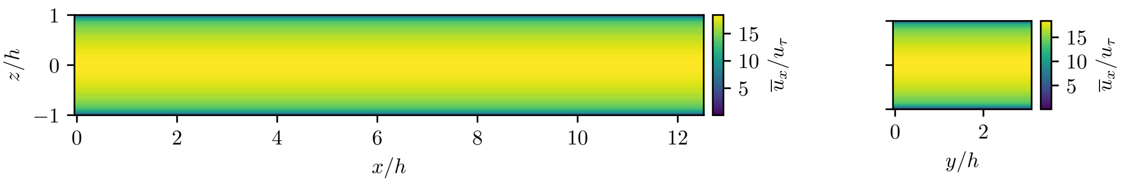


Figure 3: Accepted mean streamwise velocity $\overline{u}_x/u_\tau$ over the $60 \leq tu_\tau/h \leq 200$ target window. Left panel: the wall-normal mean replicated along x/h. Right panel: the same mean replicated along y/h.

5.3 Mean and Reynolds-stress profiles

Figure 4 shows the accepted mean velocity profile. The streamwise component has the expected channel-flow profile, while the spanwise and wall-normal mean components remain small compared with $\overline{u}_x$. The profile is plotted as $\overline{u}_i/u_\tau$ against z/h. Figure 5 shows the six independent components of the Reynolds-stress tensor profile $R_{\alpha\beta}/u_\tau^2$ against z/h. The off-diagonal component R_{xz} carries the dominant shear-stress structure; the other off-diagonal components are small on this scale.

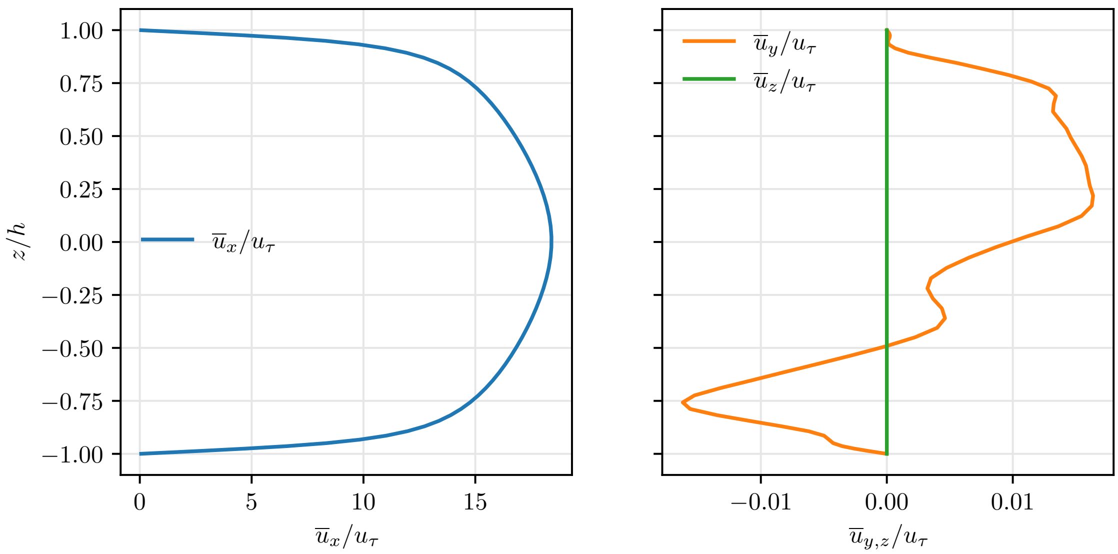


Figure 4: Accepted mean velocity profile computed from the $60 \leq tu_\tau/h \leq 200$ target window. Left panel: streamwise mean $\overline{u}_x/u_\tau$. Right panel: secondary means $\overline{u}_y/u_\tau$ and $\overline{u}_z/u_\tau$.

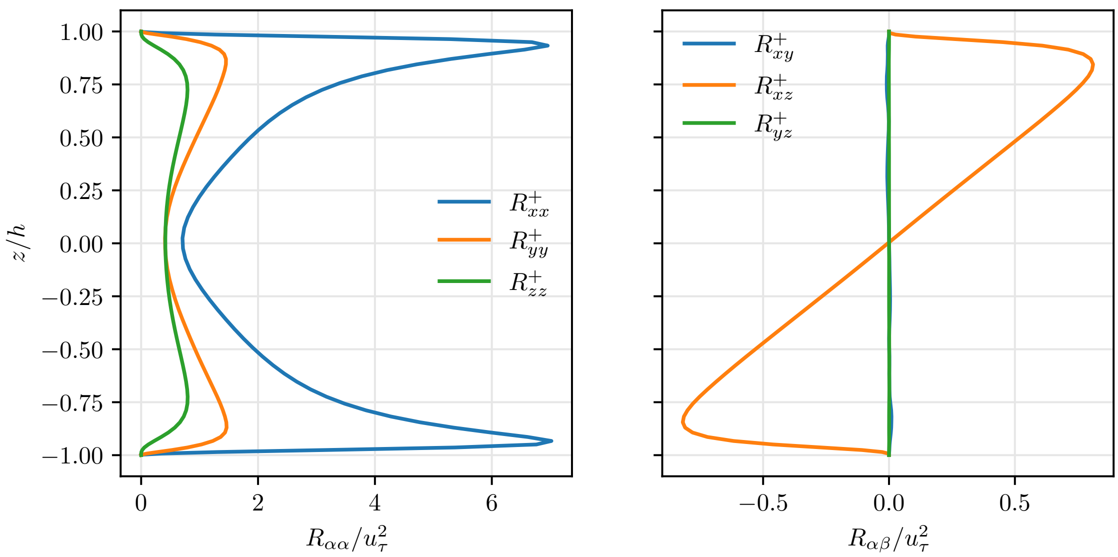


Figure 5: Accepted Reynolds-stress tensor profile computed from the $60 \leq tu_\tau/h \leq 200$ target window. Left panel: diagonal components R_{xx}/u_τ^2, R_{yy}/u_τ^2, and R_{zz}/u_τ^2. Right panel: off-diagonal components R_{xy}/u_τ^2, R_{xz}/u_τ^2, and R_{yz}/u_τ^2.

5.4 Plane-averaged temporal auto-covariances and auto-spectra

The covariance and spectral quantities in the preceding sections are equal-time statistics. For a reproduction run with explicit temporal dynamics, it is also useful to record height-resolved temporal statistics in physical space. The accepted equal-time target uses the converged production window described above. The temporal figures in this subsection instead use a separate dense continuation restarted from the final $tu_\tau/h = 200$ checkpoint and sampled every 20 DNS steps. This keeps the equal-time target unchanged while improving the temporal lag spacing and frequency range.

Let the dense sample times, measured from the first dense snapshot, be

$$t_n = n\Delta t_s, \qquad n = 0, \ldots, N_s - 1,$$

where the absolute dense-window range is

$$200.01 \leq t_{\mathrm{abs}} u_\tau/h \leq 320,$$

$N_s = 12000$, and

$$\Delta t_s u_\tau/h = 0.01.$$

The sample spacing Δt_s is the snapshot spacing, not the DNS solver step. For a velocity component $i \in \{x, y, z\}$, define

$$u_i'(x_j, y_\ell, z_m, t_n) = u_i(x_j, y_\ell, z_m, t_n) - \overline{u}_i^{\mathrm{dense}}(z_m),$$

where $\overline{u}_i^{\mathrm{dense}}(z_m)$ is recomputed from the dense continuation samples. This dense temporal mean is used only for the temporal diagnostics; the target mean profile in Eq. (16) remains the accepted $60 \leq tu_\tau/h \leq 200$ estimator. For an integer lag $s \geq 0$, let

$$\tau_s = s\Delta t_s.$$

The plane-averaged positive-lag auto-covariance at height z_m is estimated by

$$R_{ii}(\tau_s; z_m) = \frac{1}{(N_s - s)N_x N_y} \sum_{n=0}^{N_s-s-1} \sum_{j=0}^{N_x-1} \sum_{\ell=0}^{N_y-1} u_i'(x_j, y_\ell, z_m, t_{n+s}) u_i'(x_j, y_\ell, z_m, t_n). \qquad (50)$$

This is the direct finite-sample approximation of

$$\mathbb{E}\left[u_i'(t+\tau,x,y,z_m)u_i'(t,x,y,z_m)\right],$$

with the expectation replaced by an average over admissible time origins and over the statistically homogeneous x–y plane. The factor $(N_s-s)^{-1}$ appears because only N_s-s pairs are available at lag s, and the dimensionless correlation is

$$\rho_{ii}(\tau_s;z_m)=\frac{R_{ii}(\tau_s;z_m)}{R_{ii}(0;z_m)}. \tag{51}$$

Equation (51) is dimensionless and is plotted against $\tau_s u_\tau/h$.

The corresponding auto-spectrum is computed by a temporal Fourier transform at each fixed (x_j,y_ℓ,z_m), followed by the same horizontal-plane average. The angular frequencies associated with the N_s-point temporal FFT are

$$\omega_r=2\pi\,\mathrm{fftfreq}(N_s,\Delta t_s)_r. \tag{52}$$

For the dense continuation,

$$\Delta\omega^*=\frac{2\pi}{N_s\Delta t_s}\frac{h}{u_\tau}=0.0523598775599,$$

and the plotted positive branch extends to

$$\omega^*_{\max}=314.106905482,$$

with Nyquist magnitude $\pi/\Delta t_s\simeq 314.159265359$. Let w_n be an optional temporal window and define the corresponding frequency-indexed Fourier coefficient by

$$\widetilde{u}_i(x_j,y_\ell,z_m;\omega_r)=\sum_{n=0}^{N_s-1}w_n u_i'(x_j,y_\ell,z_m,t_n)\exp\left(-2\pi\mathrm{i}\frac{rn}{N_s}\right),\qquad r=0,\dots,N_s-1. \tag{53}$$

With the unnormalized discrete Fourier transform in Eq. (53), the two-sided plane-averaged periodogram is

$$S_{ii}(\omega_r;z_m)=\frac{\Delta t_s}{N_xN_y\sum_{n=0}^{N_s-1}w_n^2}\sum_{j=0}^{N_x-1}\sum_{\ell=0}^{N_y-1}\left|\widetilde{u}_i(x_j,y_\ell,z_m;\omega_r)\right|^2. \tag{54}$$

The dense auto-spectrum figure uses a Hann window. The formula records the normalization for a general window because smoothing is useful for this finite temporal record.

The normalization in Eq. (54) is chosen to be consistent with the two-sided angular-frequency convention. To see this, first consider $w_n=1$, for which $\sum_n w_n^2=N_s$. The angular-frequency spacing is

$$\Delta\omega=\frac{2\pi}{N_s\Delta t_s}.$$

Parseval's identity for the unnormalized transform gives

$$\sum_{r=0}^{N_s-1}\left|\widetilde{u}_i(x_j,y_\ell,z_m;\omega_r)\right|^2=N_s\sum_{n=0}^{N_s-1}\left|u_i'(x_j,y_\ell,z_m,t_n)\right|^2.$$

Therefore,

$$\begin{aligned}
\frac{1}{2\pi}\sum_{r=0}^{N_s-1} S_{ii}(\omega_r; z_m)\Delta\omega &= \frac{1}{N_s\Delta t_s}\sum_{r=0}^{N_s-1}\frac{\Delta t_s}{N_s N_x N_y}\sum_{j,\ell}|\widetilde{u}_i(x_j, y_\ell, z_m; \omega_r)|^2 \\
&= \frac{1}{N_s^2 N_x N_y}\sum_{j,\ell} N_s \sum_{n=0}^{N_s-1}\left|u_i'(x_j, y_\ell, z_m, t_n)\right|^2 \\
&= \frac{1}{N_s N_x N_y}\sum_{n,j,\ell}\left|u_i'(x_j, y_\ell, z_m, t_n)\right|^2 = R_{ii}(0; z_m). \qquad (55)
\end{aligned}$$

Thus, for a rectangular window, the discrete sum of the two-sided periodogram recovers the zero-lag plane-averaged variance. For a nontrivial window such as the Hann window used below, the same calculation recovers the corresponding window-energy-weighted zero-lag estimate. For Figure 7, the dimensional spectrum is normalized by its own discrete two-sided spectral area and then nondimensionalized as

$$S_{ii}^*(\omega_r; z_m) = \frac{S_{ii}(\omega_r; z_m)u_\tau}{hR_{ii}^S(0; z_m)}, \qquad \omega_r^* = \frac{\omega_r h}{u_\tau}.$$

Here $R_{ii}^S(0; z_m)$ denotes the zero-lag variance implied by the discrete spectral area. This scaling is dimensionless because $[S_{ii}/R_{ii}^S(0; z_m)] = [t]$, and it gives the normalized two-sided area identity

$$\frac{1}{2\pi}\sum_{r=0}^{N_s-1} S_{ii}^*(\omega_r; z_m)\Delta\omega^* = 1, \qquad \Delta\omega^* = \Delta\omega\, h/u_\tau.$$

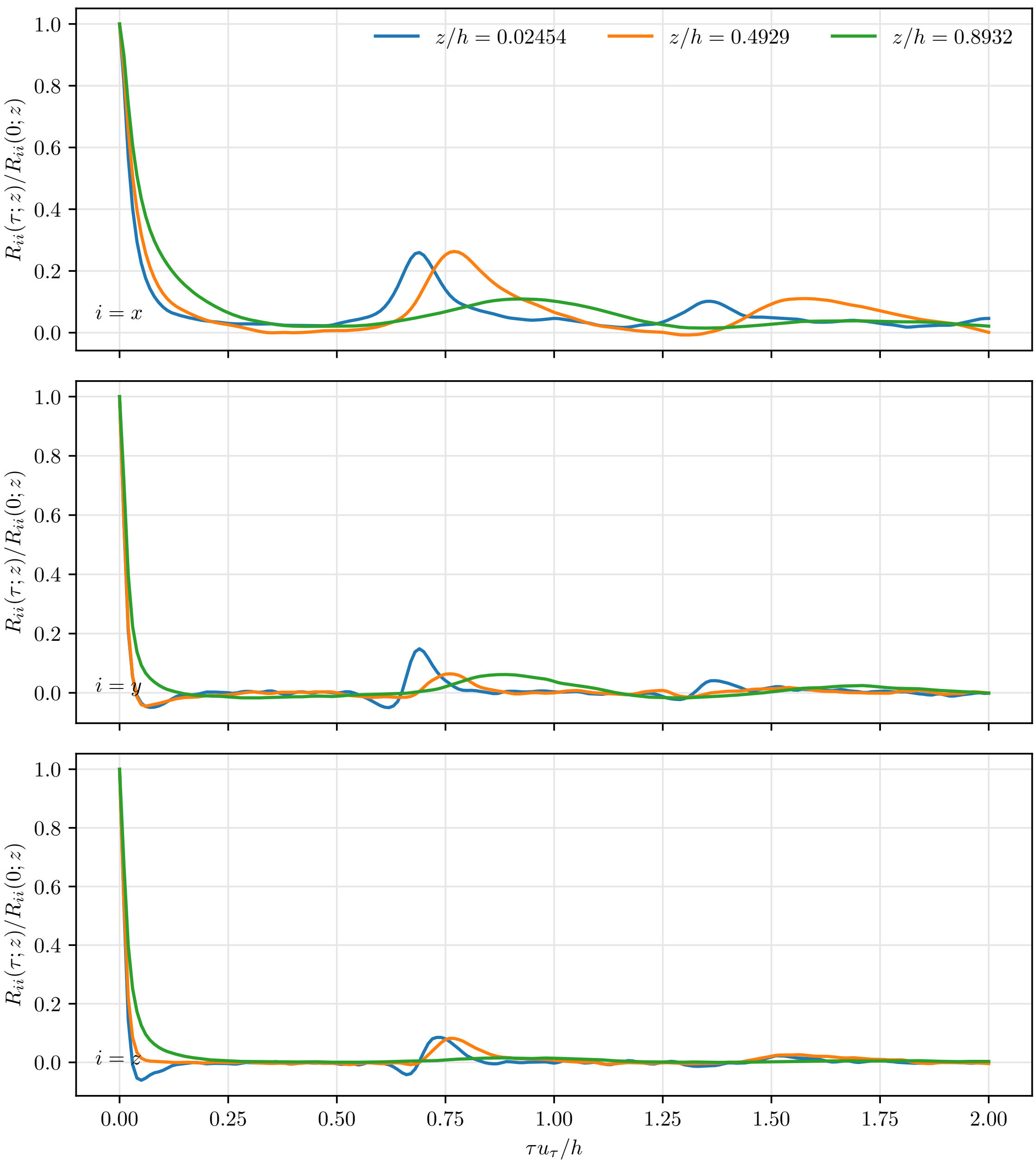


Figure 6: Plane-averaged temporal auto-correlation $R_{ii}(\tau; z)/R_{ii}(0; z)$ from the dense temporal continuation. The three panels correspond to $i = x$, $i = y$, and $i = z$, as indicated inside each panel. Curves show one nonsymmetric half-channel set of wall-normal levels, namely the nearest saved Chebyshev-Gauss levels to $z/h = 0, 0.5, 0.9$. The abscissa is the nondimensional lag $\tau u_\tau/h$, sampled at increments of 0.01.

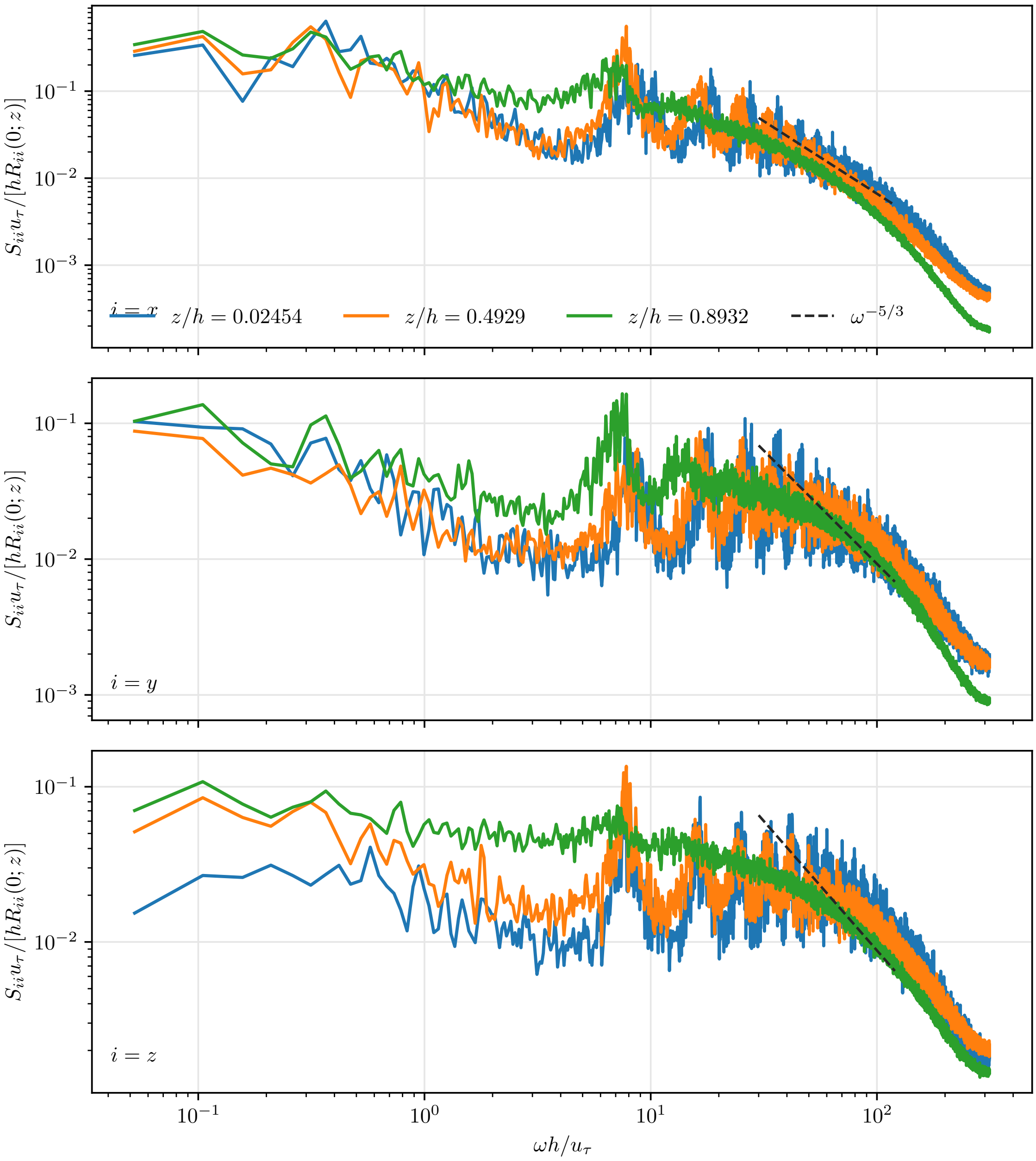


Figure 7: Plane-averaged temporal auto-spectrum computed from the dense temporal continuation using a Hann window. The plotted quantity is the variance-normalized spectrum $S_{ii}(\omega; z)u_\tau/[hR^S_{ii}(0; z)]$ against $\omega h/u_\tau$, using the positive branch of the two-sided angular-frequency periodogram. With this normalization, the two-sided spectral area equals unity for each component and wall-normal level. The three panels correspond to $i = x$, $i = y$, and $i = z$, as indicated inside each panel. Curves show one nonsymmetric half-channel set of wall-normal levels, namely the nearest saved Chebyshev-Gauss levels to $z/h = 0, 0.5, 0.9$. The dashed line gives a $\omega^{-5/3}$ reference slope.

The auto-correlation in Figure 6 shows a clear separation between the streamwise component and the two secondary components. At $\tau u_\tau/h = 0.1$, the three selected streamwise correlations are approximately 0.083, 0.127, and 0.243, ordered from the near-center level to the outer selected level. The streamwise curves then show low-amplitude recurrent peaks near $\tau u_\tau/h \simeq 0.7$–0.9, consistent with advection through the periodic streamwise domain. The spanwise and wall-normal components lose most of their correlation much sooner: the first nonpositive samples occur between 0.03 and 0.14 for five of the six selected secondary-component curves. The outer selected wall-normal curve is the exception, remaining weakly positive until approximately 1.35. Beyond the initial decay, the secondary correlations fluctuate close to zero and exhibit only weak recurrences. The one-sided wall-normal selection is used only to avoid duplicating approximately symmetric information. The dense continuation starts from the accepted production final state and is used here for temporal resolution, not as a replacement for the equal-time target.

The log-log spectra in Figure 7 give the same information in the frequency domain after removing the variance amplitude at each height. The fraction of two-sided spectral area in $|\omega|h/u_\tau \leq 5$ is approximately 12%–20% for the three streamwise curves, compared with approximately 2%–8% for the spanwise and wall-normal curves. The outer selected streamwise level has the largest low-frequency fraction. All components also show a broad band of content near $\omega h/u_\tau \simeq 7$–10, corresponding to the weak temporal recurrences visible in the correlation curves. Relative to the original $\Delta t_s u_\tau/h = 0.1$ production snapshots, the dense continuation extends the plotted positive branch by approximately one decade, to $\omega h/u_\tau \simeq 314$, and refines the lag spacing by the same factor. The dashed $\omega^{-5/3}$ segment is included only as a log-log slope reference over $\omega h/u_\tau = 30$–120; the finite temporal record and single-record periodogram should not be interpreted as an inertial-range fit. By construction, the variance-normalized two-sided spectral area is one for every plotted component and wall-normal level.

The completed dense DNS advanced from the copied $t = 200$ checkpoint to $t = 320$ in 9700.20 seconds and wrote a 75523367624-byte velocity file. The 12000 component records have identical keys from 400020 through 640000 and snapshot shape $(64, 128, 32)$. The postprocessed autocovariance and autospectrum arrays have shapes $(201, 64, 3)$ and $(12000, 64, 3)$, respectively. Their sample-time and wall-normal grids match exactly; all stored values are finite, the spectrum is nonnegative, and the maximum absolute lag-zero normalization error in the stored autocorrelation is zero in double precision.

5.5 Target layout and constraint audit

The accepted target contains the following key shapes:

$$\begin{aligned}
&\texttt{mean_profile}: && (64,3),\\
&\texttt{reynolds_stress_profile}: && (64,3,3),\\
&\texttt{modal/u_hat}: && (1401,128,32,192),\\
&\texttt{modal/B0_DNS}: && (128,32,192,192),\\
&\texttt{lag_covariance/lag_0}: && (128,32,192,192),\\
&\texttt{lag_covariance/lag_1}: && (128,32,192,192).
\end{aligned}$$

The target file size is

$$24880260208 \text{ bytes.}$$

The Chebyshev-quadrature constraint file is

$$\texttt{MKM\_constraints\_Lx4pi\_N64\_128\_32\_cheb\_quadrature\_spectral.h5}.$$

Its rank audit gives one rank-68 mode and 4095 rank-70 modes:

$$\#\{\text{rank} = 68\} = 1, \qquad \#\{\text{rank} = 70\} = 4095.$$

The final audit confirms that the target grid matches both the saved velocity grid and the saved Chebyshev-quadrature constraint grid:

$$\texttt{z\_grid\_match\_velocity} = \texttt{true}, \qquad \texttt{z\_grid\_match\_constraint} = \texttt{true}.$$

The global residual of the sampled modal covariance under the saved constraint was

$$\frac{\|\widetilde{G}\widehat{\boldsymbol{q}}\|}{\|\widehat{\boldsymbol{q}}\|} \approx 5.48 \times 10^{-14},$$

with divergence and boundary contributions

$$5.48 \times 10^{-14} \quad \text{and} \quad 3.68 \times 10^{-16},$$

respectively. These values are near machine precision for the present postprocessing and confirm consistency between the accepted target covariance and the solver-native Chebyshev/quadrature constraint.

6 Admissible projection of the sampled covariance

The preceding audit checks the sampled modal coefficients directly. In this section we also construct the covariance that would be obtained by explicitly projecting the sampled equal-time covariance onto the discrete admissible subspace. This is the covariance denoted by $B_{\mathrm{adm,DNS}}(\kappa_p, \lambda_q)$. The calculation provides an independent end-to-end check that the stored covariance, the saved constraint recipe, and the sampled Reynolds-stress profiles are mutually consistent.

6.1 Notation for one horizontal mode

Fix one horizontal Fourier mode (κ_p, λ_q), and let

$$d = 3N_z.$$

The stored equal-time DNS covariance for this mode is

$$B_{0,\mathrm{DNS}}^{pq} = \frac{1}{N_s} \sum_{n=0}^{N_s-1} \widehat{\boldsymbol{q}}_{pq}(t_n) \widehat{\boldsymbol{q}}_{pq}(t_n)^* \in \mathbb{C}^{d \times d},$$

with the same level-major component ordering as in Eq. (20). The row-orthonormal constraint matrix for the same mode is

$$\widetilde{G}_{pq} \in \mathbb{C}^{r_{pq} \times d}, \qquad \widetilde{G}_{pq} \widetilde{G}_{pq}^* = I_{r_{pq}}, \tag{56}$$

where r_{pq} is the numerical rank of $G_{\mathrm{raw}}(\kappa_p, \lambda_q)$. The admissible modal subspace is

$$\mathcal{A}_{pq} = \left\{ v \in \mathbb{C}^d : \widetilde{G}_{pq} v = 0 \right\}.$$

6.2 Derivation of the orthogonal projector

Because the rows of $\widetilde{G}_{pq}$ are orthonormal, the matrix

$$\widetilde{G}_{pq}^* \widetilde{G}_{pq}$$

is the orthogonal projector onto the row space of $\widetilde{G}_{pq}$. Indeed, for any vector $v \in \mathbb{C}^d$, the row-space component is obtained by first taking row coordinates $\widetilde{G}_{pq} v$, and then mapping those coordinates back into the physical modal-vector space with $\widetilde{G}_{pq}^*$. The complementary projector is therefore

$$P_{pq} = I_d - \widetilde{G}_{pq}^* \widetilde{G}_{pq}. \tag{57}$$

This matrix is Hermitian, since

$$P_{pq}^* = I_d - \widetilde{G}_{pq}^* \widetilde{G}_{pq} = P_{pq}.$$

It is also idempotent:

$$
\begin{aligned}
P_{pq}^2 &= \left(I_d - \widetilde{G}_{pq}^*\widetilde{G}_{pq}\right)\left(I_d - \widetilde{G}_{pq}^*\widetilde{G}_{pq}\right) \\
&= I_d - 2\widetilde{G}_{pq}^*\widetilde{G}_{pq} + \widetilde{G}_{pq}^*\left(\widetilde{G}_{pq}\widetilde{G}_{pq}^*\right)\widetilde{G}_{pq} \\
&= I_d - \widetilde{G}_{pq}^*\widetilde{G}_{pq} = P_{pq},
\end{aligned}
\tag{58}
$$

where Eq. (56) was used in the penultimate line. Finally,

$$
\widetilde{G}_{pq}P_{pq} = \widetilde{G}_{pq} - \left(\widetilde{G}_{pq}\widetilde{G}_{pq}^*\right)\widetilde{G}_{pq} = 0,
$$

so the range of P_{pq} is contained in $\mathcal{A}_{pq}$. If $v \in \mathcal{A}_{pq}$, then $\widetilde{G}_{pq}v = 0$, and $P_{pq}v = v$. Thus Eq. (57) is exactly the orthogonal projector onto the admissible subspace.

6.3 Projected covariance

Let $\widehat{\boldsymbol{q}}_{pq}$ denote a zero-mean random modal vector with covariance $B_{0,\mathrm{DNS}}^{pq}$. Its admissible projection is

$$
\widehat{\boldsymbol{q}}_{pq}^{\mathrm{adm}} = P_{pq}\widehat{\boldsymbol{q}}_{pq}.
$$

Taking the covariance of this projected vector gives

$$
\begin{aligned}
B_{\mathrm{adm,DNS}}^{pq} &= \mathbb{E}\left[\widehat{\boldsymbol{q}}_{pq}^{\mathrm{adm}}\left(\widehat{\boldsymbol{q}}_{pq}^{\mathrm{adm}}\right)^*\right] \\
&= \mathbb{E}\left[P_{pq}\widehat{\boldsymbol{q}}_{pq}\widehat{\boldsymbol{q}}_{pq}^*P_{pq}^*\right] \\
&= P_{pq}\mathbb{E}\left[\widehat{\boldsymbol{q}}_{pq}\widehat{\boldsymbol{q}}_{pq}^*\right]P_{pq}^* \\
&= P_{pq}B_{0,\mathrm{DNS}}^{pq}P_{pq},
\end{aligned}
\tag{59}
$$

because $P_{pq}^* = P_{pq}$. Equation (59) is the covariance used by a generator that enforces the discrete incompressibility and no-slip constraints exactly at the modal-covariance level.

6.4 Exact equality for constrained DNS samples

The near-zero difference between $B_{0,\mathrm{DNS}}^{pq}$ and $B_{\mathrm{adm,DNS}}^{pq}$ is not an independent empirical coincidence. It follows algebraically from using the same discrete constraint in the DNS target and in the projection. Suppose that each sampled modal coefficient satisfies the saved constraint exactly,

$$
\widetilde{G}_{pq}\widehat{\boldsymbol{q}}_{pq}(t_n) = 0, \qquad n = 0, \ldots, N_s - 1. \tag{60}
$$

Then every sample belongs to the admissible subspace $\mathcal{A}_{pq}$. Since P_{pq} is the identity on this subspace,

$$
P_{pq}\widehat{\boldsymbol{q}}_{pq}(t_n) = \widehat{\boldsymbol{q}}_{pq}(t_n). \tag{61}
$$

Applying Eq. (61) to the finite-sample covariance estimator gives

$$
\begin{aligned}
P_{pq}B_{0,\mathrm{DNS}}^{pq}P_{pq} &= P_{pq}\left[\frac{1}{N_s}\sum_{n=0}^{N_s-1}\widehat{\boldsymbol{q}}_{pq}(t_n)\widehat{\boldsymbol{q}}_{pq}(t_n)^*\right]P_{pq} \\
&= \frac{1}{N_s}\sum_{n=0}^{N_s-1}\left(P_{pq}\widehat{\boldsymbol{q}}_{pq}(t_n)\right)\left(P_{pq}\widehat{\boldsymbol{q}}_{pq}(t_n)\right)^* \\
&= \frac{1}{N_s}\sum_{n=0}^{N_s-1}\widehat{\boldsymbol{q}}_{pq}(t_n)\widehat{\boldsymbol{q}}_{pq}(t_n)^* = B_{0,\mathrm{DNS}}^{pq}.
\end{aligned}
\tag{62}
$$

Combining Eqs. (59) and (62) yields the exact identity

$$B^{pq}_{\mathrm{adm,DNS}} = B^{pq}_{0,\mathrm{DNS}} \tag{63}$$

for every horizontal mode whose sampled coefficients satisfy Eq. (60).

The same result can be written as a covariance-range statement. From Eq. (60),

$$\widetilde{G}_{pq} B^{pq}_{0,\mathrm{DNS}} = \frac{1}{N_s} \sum_{n=0}^{N_s-1} \left(\widetilde{G}_{pq} \widehat{\boldsymbol{q}}_{pq}(t_n) \right) \widehat{\boldsymbol{q}}_{pq}(t_n)^* = 0, \tag{64}$$

$$B^{pq}_{0,\mathrm{DNS}} \widetilde{G}^*_{pq} = \left(\widetilde{G}_{pq} B^{pq}_{0,\mathrm{DNS}} \right)^* = 0. \tag{65}$$

Substituting Eqs. (64) and (65) into the direct expansion of $P_{pq} B^{pq}_{0,\mathrm{DNS}} P_{pq}$ gives

$$\begin{aligned} P_{pq} B^{pq}_{0,\mathrm{DNS}} P_{pq} &= B^{pq}_{0,\mathrm{DNS}} - \widetilde{G}^*_{pq} \widetilde{G}_{pq} B^{pq}_{0,\mathrm{DNS}} - B^{pq}_{0,\mathrm{DNS}} \widetilde{G}^*_{pq} \widetilde{G}_{pq} + \widetilde{G}^*_{pq} \widetilde{G}_{pq} B^{pq}_{0,\mathrm{DNS}} \widetilde{G}^*_{pq} \widetilde{G}_{pq} \\ &= B^{pq}_{0,\mathrm{DNS}}. \end{aligned} \tag{66}$$

Hence the projection cannot change a covariance assembled from exactly constrained samples. Any nonzero $\|B^{pq}_{\mathrm{adm,DNS}} - B^{pq}_{0,\mathrm{DNS}}\|$ in the production calculation is therefore a finite-precision diagnostic: it measures floating-point roundoff, SVD compression and reconstruction tolerances, and HDF5 read/write arithmetic, not a physical or statistical modification of the target covariance.

The implementation evaluates Eq. (59) from the stored data. If a dense $\widetilde{G}_{pq}$ is present in the constraint file, it is read directly. The production constraint file stores seven representative compressed matrices by default; for the remaining modes, the postprocessor reconstructs $\widetilde{G}_{pq}$ from the saved arrays D_z, B_∂, E_x, E_y, E_z, κ_p, and λ_q by applying the same SVD compression derived in Section 4. Thus all ingredients are taken from the target and constraint HDF5 files. The numerical formula used in the script is the expanded form

$$\begin{aligned} B^{pq}_{\mathrm{adm,DNS}} = B^{pq}_{0,\mathrm{DNS}} &- \widetilde{G}^*_{pq} \widetilde{G}_{pq} B^{pq}_{0,\mathrm{DNS}} - B^{pq}_{0,\mathrm{DNS}} \widetilde{G}^*_{pq} \widetilde{G}_{pq} \\ &+ \widetilde{G}^*_{pq} \widetilde{G}_{pq} B^{pq}_{0,\mathrm{DNS}} \widetilde{G}^*_{pq} \widetilde{G}_{pq}, \end{aligned} \tag{67}$$

followed by explicit Hermitian symmetrization at roundoff level.

6.5 Reynolds stresses reproduced from a modal covariance

The sampled Reynolds-stress profile can also be reconstructed from any modal covariance C^{pq} with the same Fourier normalization as the target file. Let $s(x) = 0$, $s(y) = 1$, and $s(z) = 2$, and define the level-major index

$$\ell(m, \alpha) = 3m + s(\alpha).$$

The inverse horizontal Fourier representation of the fluctuation field is

$$u'_\alpha(x_j, y_\ell, z_m) = \sum_{p=0}^{N_x-1} \sum_{q=0}^{N_y-1} \widehat{u}_\alpha(z_m; \kappa_p, \lambda_q) \exp\left[\mathrm{i}(\kappa_p x_j + \lambda_q y_\ell)\right],$$

because the forward transform in Eq. (19) already contains the factor $1/(N_x N_y)$. Multiplying two components and averaging over the horizontal grid gives

$$\left\langle u'_\alpha u'_\beta \right\rangle_{xy} = \sum_{p,q} \sum_{p',q'} \widehat{u}_\alpha(z_m; \kappa_p, \lambda_q) \widehat{u}_\beta(z_m; \kappa_{p'}, \lambda_{q'})^* \left\langle e^{\mathrm{i}[(\kappa_p - \kappa_{p'})x + (\lambda_q - \lambda_{q'})y]} \right\rangle_{xy}. \tag{68}$$

The discrete Fourier modes are orthogonal on the sampled periodic grid, so

$$\left\langle e^{\mathrm{i}[(\kappa_p-\kappa_{p'})x+(\lambda_q-\lambda_{q'})y]}\right\rangle_{xy} = \delta_{pp'}\delta_{qq'}.$$

Therefore only equal horizontal modes survive, and the Reynolds-stress profile implied by C^{pq} is

$$R^{C}_{\alpha\beta}(z_m) = \sum_{p=0}^{N_x-1}\sum_{q=0}^{N_y-1} C^{pq}_{\ell(m,\alpha),\ell(m,\beta)}. \tag{69}$$

No additional N_xN_y factor appears in Eq. (69); it has already been fixed by the forward and inverse transform convention. In the comparison below, Eq. (69) is evaluated for both $C^{pq} = B^{pq}_{0,\mathrm{DNS}}$ and $C^{pq} = B^{pq}_{\mathrm{adm,DNS}}$.

6.6 Production computation and comparison

The projected covariance was computed with

`scripts/mkm_dns_target/project_mkm_dns_covariance.py`.

For the production data set, the script wrote

`MKM_production_Lx4pi_64_128_32_projected_covariance_t60_t200.h5`,

which contains $B^{pq}_{\mathrm{adm,DNS}}$, the Reynolds profiles implied by $B_{0,\mathrm{DNS}}$ and $B_{\mathrm{adm,DNS}}$, and modewise projection diagnostics. The run processed all $128\times 32 = 4096$ horizontal modes. Seven $\widetilde{G}_{pq}$ matrices were read directly from saved representative-mode groups, and the remaining 4089 were reconstructed from the saved constraint operators.

Table 2: Projection and Reynolds-stress consistency diagnostics for the accepted $60 \le tu_\tau/h \le 200$ production target. Norms are Frobenius norms unless otherwise stated.

Diagnostic	Value
Global $\Vert B_{\mathrm{adm,DNS}} - B_{0,\mathrm{DNS}}\Vert/\Vert B_{0,\mathrm{DNS}}\Vert$	1.0967×10^{-15}
Maximum modewise $\Vert B^{pq}_{\mathrm{adm,DNS}} - B^{pq}_{0,\mathrm{DNS}}\Vert/\Vert B^{pq}_{0,\mathrm{DNS}}\Vert$	2.1537×10^{-15}
Relative trace change $(\mathrm{tr}\,B_{\mathrm{adm,DNS}} - \mathrm{tr}\,B_{0,\mathrm{DNS}})/\mathrm{tr}\,B_{0,\mathrm{DNS}}$	-2.2079×10^{-16}
Total $\mathrm{tr}\,B_{0,\mathrm{DNS}}$	257.4582475
Maximum projected constraint residual	5.4182×10^{-17}
$\Vert R^{B_{0,\mathrm{DNS}}} - R^{\mathrm{sample}}\Vert/\Vert R^{\mathrm{sample}}\Vert$	2.4487×10^{-15}
$\Vert R^{B_{\mathrm{adm,DNS}}} - R^{\mathrm{sample}}\Vert/\Vert R^{\mathrm{sample}}\Vert$	2.4532×10^{-15}
$\Vert R^{B_{\mathrm{adm,DNS}}} - R^{B_{0,\mathrm{DNS}}}\Vert/\Vert R^{B_{0,\mathrm{DNS}}}\Vert$	2.1223×10^{-16}
$\max \lvert R^{B_{\mathrm{adm,DNS}}} - R^{\mathrm{sample}}\rvert$	2.9310×10^{-14}

The values in Table 2 are therefore best interpreted as a finite-precision verification of Eq. (63). The exact-arithmetic prediction is that the explicit projection leaves the production equal-time covariance unchanged. The observed 10^{-15}-level changes are consistent with the direct modal-coefficient audit in Section 5: the sampled velocity coefficients already satisfy the saved Chebyshev/quadrature divergence and no-slip constraints to near machine precision.

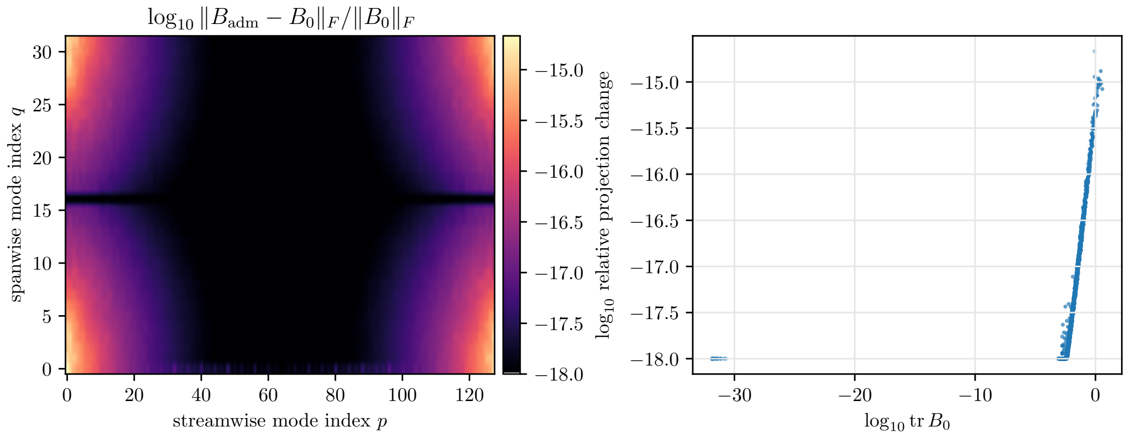


Figure 8: Modewise finite-precision effect of the admissible covariance projection. Left: relative Frobenius-norm change $\|B^{pq}_{\mathrm{adm,DNS}} - B^{pq}_{0,\mathrm{DNS}}\|_F/\|B^{pq}_{0,\mathrm{DNS}}\|_F$. Right: the same relative projection change plotted against modal covariance energy $\mathrm{tr}\,B^{pq}_{0,\mathrm{DNS}}$.

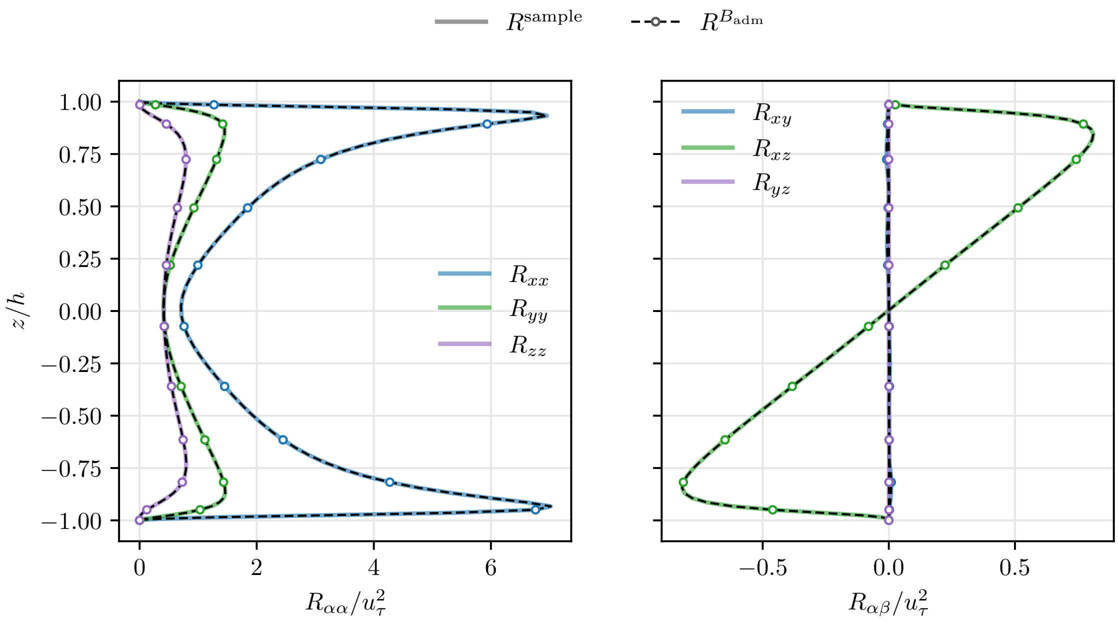


Figure 9: Reynolds-stress profiles from the sampled physical-space estimator and from the projected modal covariance. Solid colored curves show R^{sample}. The projected profiles $R^{B_{\mathrm{adm,DNS}}}$, computed from Eq. (69), are drawn as black dashed curves with open colored markers so that the exact overlap is visible.

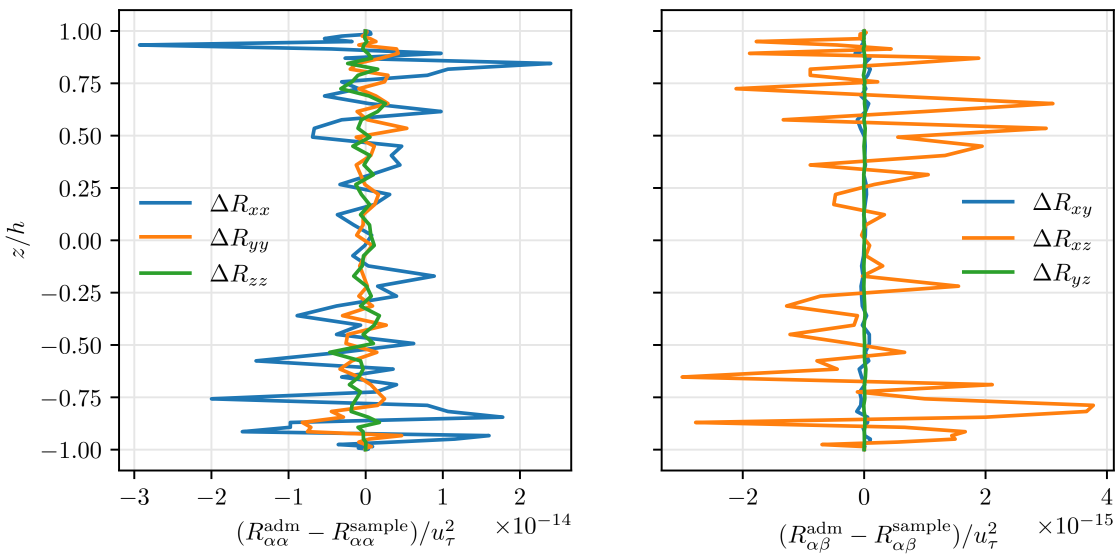


Figure 10: Pointwise differences between the Reynolds-stress profiles implied by $B_{\rm adm,DNS}$ and those sampled directly in physical space. The horizontal axes are nondimensionalized by u_τ^2. The errors are at roundoff level relative to the profile magnitudes.

7 Selected-mode channel resolvent analysis

The constraint construction in Section 4 and the modal DNS objects in Sections 3 and 6 provide the ingredients needed for a McKeon-style channel resolvent calculation. This section records the implemented proof-of-concept framework. The implementation is selected-mode: it computes the resolvent, DNS temporal cross-spectral density (CSD), DNS projection fractions, and diagnostic figures for specified horizontal mode indices. It is not yet an exhaustive mode sweep and should not be interpreted as a high-Reynolds-number or long-box scaling result.

7.1 Notation for a single horizontal mode

Fix a horizontal Fourier mode with indices (p, q), streamwise wavenumber $\kappa = \kappa_p$, and spanwise wavenumber $\lambda = \lambda_q$. For this mode the level-major velocity vector is

$$\boldsymbol{q} = \begin{bmatrix} u_x(z_0) & u_y(z_0) & u_z(z_0) & \cdots & u_x(z_{N_z-1}) & u_y(z_{N_z-1}) & u_z(z_{N_z-1}) \end{bmatrix}^T \in \mathbb{C}^{3N_z}. \tag{70}$$

The component ordering in Eq. (70) is the same ordering used by the target HDF5 file and by the constraint matrices: streamwise, spanwise, wall-normal at each Chebyshev-Gauss level.

The turbulent mean profile is the streamwise component of the sampled mean,

$$U_m = U(z_m) = (\texttt{mean\_profile})_{m,0}\,, \qquad m = 0, \ldots, N_z - 1. \tag{71}$$

The derivative matrix D_z derived in Section 4 gives the discrete wall-normal mean shear

$$U'_m = (D_z U)_m\,. \tag{72}$$

All vectors and matrices below use the Chebyshev-Gauss physical quadrature weights, not a uniform-grid approximation. Let $w_m > 0$ denote the Fejer first-rule weights on $[-1, 1]$, ordered consistently with z_m. They are defined by the quadrature rule

$$\int_{-1}^{1} g(z)\,\mathrm{d}z \approx \sum_{m=0}^{N_z-1} w_m g(z_m), \qquad \sum_{m=0}^{N_z-1} w_m \approx 2. \tag{73}$$

The implementation stores no kinetic-energy factor 1/2 in the weight. The level-major velocity energy matrix is therefore

$$W_q = \mathrm{diag}\left(w_0, w_0, w_0, \ldots, w_{N_z-1}, w_{N_z-1}, w_{N_z-1}\right). \tag{74}$$

For two modal velocity vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}^{3N_z}$, the implemented energy inner product is

$$\langle \boldsymbol{a}, \boldsymbol{b} \rangle_{W_q} = \boldsymbol{a}^* W_q \boldsymbol{b} = \sum_{m=0}^{N_z-1} w_m \left[a_x(z_m)^* b_x(z_m) + a_y(z_m)^* b_y(z_m) + a_z(z_m)^* b_z(z_m) \right]. \tag{75}$$

7.2 Energy-orthonormal admissible basis

For the chosen (κ, λ), the compressed constraint matrix $\widetilde{G}(\kappa, \lambda)$ was derived in Section 4. The admissible velocity subspace is

$$\mathcal{V}_{\kappa,\lambda} = \left\{ \boldsymbol{q} \in \mathbb{C}^{3N_z} : \widetilde{G}(\kappa, \lambda) \boldsymbol{q} = \boldsymbol{0} \right\}. \tag{76}$$

The same singular-value convention used to construct $\widetilde{G}$ also gives a Euclidean null-space matrix N, denoted `Nmat` in the code, such that

$$\widetilde{G} N = 0. \tag{77}$$

The columns of N span the admissible space, but they are orthonormal in the Euclidean inner product inherited from the SVD, not in the physical quadrature energy inner product. The implemented resolvent instead needs a matrix Q satisfying

$$\widetilde{G} Q = 0, \qquad Q^* W_q Q = I. \tag{78}$$

The construction is as follows. First form the Gram matrix

$$M = N^* W_q N. \tag{79}$$

Because W_q is positive diagonal and the columns of N are linearly independent, M is Hermitian positive definite on the retained null space. Let

$$M = V \Lambda V^* \tag{80}$$

be its Hermitian eigendecomposition, where $\Lambda = \mathrm{diag}(\mu_1, \ldots, \mu_r)$ with $\mu_j > 0$, and r is the admissible dimension. Define

$$Q = N V \Lambda^{-1/2}. \tag{81}$$

Then the constraint property follows immediately from Eq. (77):

$$\widetilde{G} Q = \widetilde{G} N V \Lambda^{-1/2} = 0. \tag{82}$$

The energy orthonormality follows by substituting Eq. (81) into $Q^* W_q Q$:

$$\begin{aligned} Q^* W_q Q &= \left(N V \Lambda^{-1/2} \right)^* W_q \left(N V \Lambda^{-1/2} \right) \\ &= \Lambda^{-1/2} V^* N^* W_q N V \Lambda^{-1/2} \\ &= \Lambda^{-1/2} V^* M V \Lambda^{-1/2} \\ &= \Lambda^{-1/2} \Lambda \Lambda^{-1/2} = I. \end{aligned} \tag{83}$$

Consequently every admissible velocity vector represented by the reduced coordinate vector $\boldsymbol{a} \in \mathbb{C}^r$ has the form

$$\boldsymbol{q} = Q \boldsymbol{a}, \tag{84}$$

and its quadrature energy is

$$\boldsymbol{q}^* W_q \boldsymbol{q} = \boldsymbol{a}^* Q^* W_q Q \boldsymbol{a} = \boldsymbol{a}^* \boldsymbol{a}. \tag{85}$$

This identity is why the singular vectors of the reduced resolvent can be mapped back to physical velocity space without any additional normalization.

7.3 Pressure-eliminated velocity operator

The velocity perturbation is written as a single Fourier harmonic,

$$\boldsymbol{u}'(x,y,z,t) = \widehat{\boldsymbol{u}}(z,t)\exp\left[\mathrm{i}(\kappa x + \lambda y)\right], \tag{86}$$

and the mean velocity is $U(z)\boldsymbol{e}_x$. Linearizing Eq. (4) around this mean gives

$$\partial_t \boldsymbol{u}' = -U\partial_x \boldsymbol{u}' - u_z' U' \boldsymbol{e}_x - \nabla p' + \frac{1}{Re_\tau}\Delta \boldsymbol{u}' + \boldsymbol{f}', \tag{87}$$

where $\boldsymbol{f}'$ is a generic harmonic forcing representing the nonlinear forcing residual in a resolvent interpretation. Equation (87) follows term by term from $(U\boldsymbol{e}_x + \boldsymbol{u}')\cdot\nabla(U\boldsymbol{e}_x + \boldsymbol{u}')$: retaining only terms linear in $\boldsymbol{u}'$ gives $U\partial_x\boldsymbol{u}'$ from advection of the perturbation by the mean and $u_z'U'\boldsymbol{e}_x$ from wall-normal displacement of the mean profile.

Substituting the spatial harmonic in Eq. (86) gives

$$\partial_x \mapsto \mathrm{i}\kappa, \qquad \partial_y \mapsto \mathrm{i}\lambda, \qquad \Delta \mapsto \partial_{zz} - (\kappa^2 + \lambda^2).$$

The wall-normal second derivative is represented by

$$D_2 = D_z D_z. \tag{88}$$

If $\alpha^2 = \kappa^2 + \lambda^2$, the pressure-free part of the discrete linearized velocity operator is therefore

$$L_{\mathrm{raw}}\boldsymbol{q} = -\mathrm{i}\kappa U \boldsymbol{q} - q_z U' \boldsymbol{e}_x + \frac{1}{Re_\tau}\left(D_2 - \alpha^2 I\right)\boldsymbol{q}. \tag{89}$$

The notation in Eq. (89) is level-major: the first term multiplies all three velocity components at each z_m by $-\mathrm{i}\kappa U_m$, the lift-up term adds $-U_m' q_z(z_m)$ only to the streamwise component, and the viscous scalar operator $D_2 - \alpha^2 I$ is applied independently to each velocity component.

The pressure term does no work against exactly divergence-free no-slip test functions. In continuous form, if $\boldsymbol{v}$ is incompressible and satisfies $\boldsymbol{v} = \boldsymbol{0}$ at the walls and periodicity in x, y, then

$$\int_\Omega \boldsymbol{v}^* \cdot \nabla p \,\mathrm{d}\Omega = \int_{\partial\Omega} p\, \boldsymbol{v}^* \cdot \boldsymbol{n}\,\mathrm{d}S - \int_\Omega p\,\nabla\cdot\boldsymbol{v}^*\,\mathrm{d}\Omega = 0. \tag{90}$$

The implemented proof of concept uses the corresponding velocity-only admissible-subspace formulation: construct L_{raw} from Eq. (89), restrict both trial and test vectors to the columns of Q, and form

$$A = Q^* W_q L_{\mathrm{raw}} Q. \tag{91}$$

The matrix $A \in \mathbb{C}^{r\times r}$ is the reduced generator used by the single-mode resolvent solver.

7.4 Resolvent sign convention and singular modes

Let $\boldsymbol{a}(t)$ be the reduced velocity coordinate and let $\boldsymbol{b}(t)$ be the reduced forcing coordinate. With the projection above,

$$\frac{\mathrm{d}\boldsymbol{a}}{\mathrm{d}t} = A\boldsymbol{a} + \boldsymbol{b}. \tag{92}$$

The implemented Fourier-time convention is

$$\boldsymbol{a}(t) = \widehat{\boldsymbol{a}}\exp(-\mathrm{i}\omega t), \qquad \boldsymbol{b}(t) = \widehat{\boldsymbol{b}}\exp(-\mathrm{i}\omega t). \tag{93}$$

Substituting Eq. (93) into Eq. (92) gives

$$-\mathrm{i}\omega\widehat{\boldsymbol{a}} = A\widehat{\boldsymbol{a}} + \widehat{\boldsymbol{b}}. \tag{94}$$

Moving the term $A\widehat{\boldsymbol{a}}$ to the left gives

$$(-\mathrm{i}\omega I - A)\,\widehat{\boldsymbol{a}} = \widehat{\boldsymbol{b}}. \tag{95}$$

Thus the reduced resolvent matrix is

$$H(\omega) = (-\mathrm{i}\omega I - A)^{-1}, \qquad \widehat{\boldsymbol{a}} = H(\omega)\widehat{\boldsymbol{b}}. \tag{96}$$

The singular-value decomposition computed by the code is

$$H(\omega) = \Psi(\omega)\Sigma(\omega)\Phi(\omega)^*, \tag{97}$$

where the diagonal entries of Σ are nonnegative and sorted in nonincreasing order. The reduced response modes are the columns of Ψ, and the reduced forcing modes are the columns of Φ. The physical velocity modes written to HDF5 are

$$\psi_j^{\mathrm{phys}} = Q\Psi_j, \qquad \phi_j^{\mathrm{phys}} = Q\Phi_j. \tag{98}$$

Because of Eq. (83) and the Euclidean orthonormality of singular vectors, these physical modes satisfy

$$\left(\psi_j^{\mathrm{phys}}\right)^* W_q \psi_k^{\mathrm{phys}} = \delta_{jk}, \qquad \left(\phi_j^{\mathrm{phys}}\right)^* W_q \phi_k^{\mathrm{phys}} = \delta_{jk}. \tag{99}$$

The wall-normal energy-density diagnostic for a response mode is

$$e_j(z_m;\omega) = \left|\psi_{j,x}^{\mathrm{phys}}(z_m;\omega)\right|^2 + \left|\psi_{j,y}^{\mathrm{phys}}(z_m;\omega)\right|^2 + \left|\psi_{j,z}^{\mathrm{phys}}(z_m;\omega)\right|^2. \tag{100}$$

The component densities are the three terms in Eq. (100). The integrated mode energy is $\sum_m w_m e_j(z_m;\omega)$, which equals one up to roundoff for stored response modes.

For $\kappa \neq 0$, the phase speed is

$$c = \frac{\omega}{\kappa}. \tag{101}$$

Critical-layer locations are roots of

$$U(z_c) = c. \tag{102}$$

The implementation sorts the Chebyshev-Gauss nodes by z, searches for sign changes in $U(z) - c$, and linearly interpolates each crossing. If $\kappa = 0$, Eq. (101) has no finite phase speed, so the critical-layer table is deliberately empty.

7.5 Mode-resolved DNS temporal CSD

For comparison with the resolvent response modes, the workflow estimates a temporal CSD matrix for selected horizontal modes. For one selected mode, let $\boldsymbol{q}_n \in \mathbb{C}^{3N_z}$ be the modal DNS velocity fluctuation at sample time t_n. The series can come directly from the target dataset `modal/u_hat`, or it can be computed on the fly from raw velocity snapshots by subtracting the target mean profile and taking only the requested horizontal Fourier coefficients.

If temporal demeaning is requested, the selected-record mean is removed before segmentation:

$$\boldsymbol{q}_n \leftarrow \boldsymbol{q}_n - \frac{1}{N_t}\sum_{\ell=0}^{N_t-1} \boldsymbol{q}_\ell. \tag{103}$$

Let a segment have length L, sample spacing Δt_s, start index s, and temporal window g_n, $n = 0, \ldots, L-1$. The unnormalized temporal DFT used by `numpy.fft.fft` is

$$\widehat{\boldsymbol{q}}_k^{(s)} = \sum_{n=0}^{L-1} g_n \boldsymbol{q}_{s+n} \exp\left(-2\pi\mathrm{i}kn/L\right), \qquad k = 0, \ldots, L-1. \tag{104}$$

The corresponding two-sided angular frequencies are

$$\omega_k = 2\pi\,\mathrm{fftfreq}(L, \Delta t_s)_k, \tag{105}$$

and are stored sorted in ascending order. If there are N_{seg} segments and

$$E_g = \sum_{n=0}^{L-1} g_n^2, \tag{106}$$

then the implemented CSD estimator is

$$S_{qq}(\omega_k) = \frac{\Delta t_s}{E_g N_{\mathrm{seg}}} \sum_{s=1}^{N_{\mathrm{seg}}} \widehat{\boldsymbol{q}}_k^{(s)} \left(\widehat{\boldsymbol{q}}_k^{(s)}\right)^* . \tag{107}$$

This normalization is chosen so that Parseval's identity has the expected weighted-energy form. With the unnormalized DFT in Eq. (104),

$$\sum_{k=0}^{L-1} \left(\widehat{\boldsymbol{q}}_k^{(s)}\right)^* W_q \widehat{\boldsymbol{q}}_k^{(s)} = L \sum_{n=0}^{L-1} g_n^2 \boldsymbol{q}_{s+n}^* W_q \boldsymbol{q}_{s+n}. \tag{108}$$

The frequency spacing is

$$\Delta\omega = \frac{2\pi}{L\Delta t_s}. \tag{109}$$

Using Eqs. (107) and (109),

$$\begin{aligned}
\frac{1}{2\pi}\sum_{k=0}^{L-1} \mathrm{tr}(W_q S_{qq}(\omega_k))\,\Delta\omega &= \frac{1}{2\pi}\sum_{k=0}^{L-1} \frac{\Delta t_s}{E_g N_{\mathrm{seg}}} \sum_s \left(\widehat{\boldsymbol{q}}_k^{(s)}\right)^* W_q \widehat{\boldsymbol{q}}_k^{(s)} \frac{2\pi}{L\Delta t_s} \\
&= \frac{1}{E_g N_{\mathrm{seg}}} \sum_s \frac{1}{L} \sum_{k=0}^{L-1} \left(\widehat{\boldsymbol{q}}_k^{(s)}\right)^* W_q \widehat{\boldsymbol{q}}_k^{(s)} \\
&= \frac{1}{E_g N_{\mathrm{seg}}} \sum_s \sum_{n=0}^{L-1} g_n^2 \boldsymbol{q}_{s+n}^* W_q \boldsymbol{q}_{s+n}.
\end{aligned} \tag{110}$$

Thus the integrated CSD energy equals the window-weighted segment-average modal energy. For a rectangular window and one full-record segment, this is the ordinary selected-record time-mean modal energy. For Hann-windowed, overlapping production estimates, Eq. (110) is still the diagnostic being checked, but it is a window-weighted segment-average identity rather than an unwindowed whole-record variance.

7.6 Projection of DNS CSD onto response modes

At a frequency where both the resolvent and DNS CSD are available, let

$$\Psi_r = \begin{bmatrix} \psi_1^{\mathrm{phys}} & \cdots & \psi_r^{\mathrm{phys}} \end{bmatrix} \in \mathbb{C}^{3N_z \times r}$$

be the first r stored response modes. The columns satisfy $\Psi_r^* W_q \Psi_r = I_r$. The total DNS spectral energy at that frequency is

$$E_{\mathrm{DNS}}(\omega) = \mathrm{tr}(W_q S_{qq}(\omega)) . \tag{111}$$

The response-basis coefficient matrix is

$$C_r(\omega) = \Psi_r^* W_q S_{qq}(\omega) W_q \Psi_r. \tag{112}$$

To see why Eq. (112) is the correct coefficient matrix, define the rank-r CSD reconstruction

$$S_r(\omega) = \Psi_r C_r(\omega) \Psi_r^*. \tag{113}$$

Taking its weighted trace and using cyclic invariance gives

$$\mathrm{tr}(W_q S_r) = \mathrm{tr}(W_q \Psi_r C_r \Psi_r^*) = \mathrm{tr}(\Psi_r^* W_q \Psi_r C_r) = \mathrm{tr}(C_r)\,, \tag{114}$$

because $\Psi_r^* W_q \Psi_r = I_r$. Therefore the DNS energy associated with the j-th response direction is the j-th diagonal entry of C_r:

$$E_j(\omega) = (C_r(\omega))_{jj} = \left(\psi_j^{\mathrm{phys}}\right)^* W_q S_{qq}(\omega) W_q \psi_j^{\mathrm{phys}}. \tag{115}$$

The rank-one and cumulative rank-r fractions reported by the code are

$$F_j(\omega) = \frac{\mathrm{Re}\, E_j(\omega)}{E_{\mathrm{DNS}}(\omega)}, \qquad F_{\le r}(\omega) = \frac{\mathrm{Re}\,\mathrm{tr}\, C_r(\omega)}{E_{\mathrm{DNS}}(\omega)}. \tag{116}$$

The implementation also reports the weighted Frobenius relative error of Eq. (113),

$$\epsilon_r(\omega) = \frac{\left\| W_q^{1/2} \left[S_{qq}(\omega) - S_r(\omega)\right] W_q^{1/2} \right\|_F}{\left\| W_q^{1/2} S_{qq}(\omega) W_q^{1/2} \right\|_F}. \tag{117}$$

Before projection, frequencies are matched exactly to the selected CSD grid within a prescribed tolerance; a mismatch is treated as an error rather than interpolated silently.

7.7 Implemented scripts and HDF5 outputs

The framework is implemented by the scripts listed in Table 3.

Table 3: Main scripts in the selected-mode channel-resolvent workflow.

Script	Role
`mkm_channel_resolvent_utils.py`	Shared constraints, weights, bases, operators, and diagnostics.
`compute_mkm_channel_resolvent.py`	Single-mode resolvent HDF5 writer.
`plot_mkm_channel_resolvent.py`	Mode-shape, gain, critical-layer, and reconstructed-field PDFs.
`compute_mkm_modal_csd.py`	Selected-mode temporal DNS CSD estimator.
`project_mkm_dns_onto_resolvent.py`	DNS CSD projection fractions.
`run_mkm_channel_resolvent_selected_modes.py`	Selected-mode workflow driver.
`run_mkm_channel_resolvent_production_selected_modes.py`	Production wrapper with server paths and missing-file checks.
`report_mkm_channel_resolvent_workflow.py`	Markdown audit report generator.

The single-mode resolvent HDF5 file contains the groups in Table 4. The CSD and projection files contain the groups in Tables 5 and 6.

Table 4: Main HDF5 datasets written by the single-mode resolvent solver.

Dataset	Meaning
`geometry/z_wall`, `geometry/k_stream`, `geometry/k_span`	Wall-normal grid and horizontal wavenumber arrays.
`mean/U`, `mean/Uprime`	Turbulent mean profile and $U' = D_z U$.
`mode/index`, `mode/kappa`, `mode/lambda`	Selected mode indices and physical wavenumbers.
`frequencies/omega`	Stored angular frequencies.
`critical_layers/z`, `critical_layers/y_plus_nearest_wall`, `critical_layers/count`	Roots of $U(z) = \omega/\kappa$, corresponding nearest-wall y^+, and root counts.
`resolvent/singular_values`	Diagonal entries of $\Sigma(\omega)$.
`resolvent/response_modes`, `resolvent/forcing_modes`	W_q-normalized physical modes $Q\Psi_j$ and $Q\Phi_j$.
`resolvent/response_energy_density`, `resolvent/component_energy_density`	Unweighted pointwise densities $e_j(z_m)$ and component contributions.
`diagnostics/*`	Constraint residuals, energy-norm errors, basis orthonormality error, resolvent condition numbers, and constraint singular values.

Table 5: Main HDF5 datasets written by the selected-mode CSD estimator.

Dataset	Meaning
`mode/index`, `mode/k_stream`, `mode/k_span`	Selected mode rows and physical wavenumbers.
`frequencies/omega`	Sorted two-sided angular frequency grid.
`csd/Sqq`	S_{qq}, shape $(N_{\mathrm{mode}}, N_\omega, 3N_z, 3N_z)$.
`csd/trace`	Ordinary matrix trace of S_{qq}.
`csd/component_trace`	Wall-normal-weighted component energy spectra.
`csd/energy_trace`	$\mathrm{tr}(W_q S_{qq})$.
`metadata/*`	Selected times, snapshot keys, window, window energy, sample spacing, segment length, overlap, source, and demeaning flag.
`diagnostics/parseval_energy_time`, `diagnostics/parseval_energy_spectrum`, `diagnostics/parseval_relative_error`	The two sides of Eq. (110) and their relative difference.

Table 6: Main HDF5 datasets written by the DNS-to-resolvent projection.

Dataset	Meaning
`mode/index`, `mode/k_stream`, `mode/k_span`	Matched single resolvent mode.
`frequencies/resolvent_omega`, `frequencies/csd_omega`, `frequencies/csd_index`	Resolvent frequencies, matched CSD frequencies, and CSD grid indices.
`projection/energy_total`	$E_{\mathrm{DNS}} = \mathrm{tr}(W_q S_{qq})$.
`projection/energy_fraction`	F_j in Eq. (116).
`projection/cumulative_energy_fraction`	$F_{\leq r}$ for ranks $1, \ldots, r_{\max}$.
`projection/modal_coefficients`	$C_r = \Psi_r^* W_q S_{qq} W_q \Psi_r$.
`projection/weighted_frobenius_relative_error`	ϵ_r from Eq. (117).
`diagnostics/*`	Frequency match errors, mode match error, response norm errors, and negative-fraction counts.

The plotting script reads the physical response and forcing modes and writes four PDFs for each selected mode:

$$\begin{aligned} &\texttt{mkm\_resolvent\_mode\_shapes.pdf}, \\ &\texttt{mkm\_resolvent\_gain\_bode.pdf}, \\ &\texttt{mkm\_resolvent\_peak\_location\_gain.pdf}, \\ &\texttt{mkm\_resolvent\_reconstructed\_fields.pdf}. \end{aligned} \tag{118}$$

The reconstructed fields use

$$u_\alpha^{\mathrm{real}}(x, y, z) = \mathrm{Re}\left[\psi_\alpha(z) \exp\left(\mathrm{i}(\kappa x + \lambda y + \varphi)\right)\right], \qquad \alpha \in \{x, y, z\}, \tag{119}$$

where φ is the requested plotting phase.

7.8 Production selected-mode runs

Two production selected-mode workflows use the accepted long-box target and its matching Chebyshev/quadrature constraint file. The selected modes were

$$(p, q) = (1, 0), \quad (1, 1), \quad (2, 0), \quad (2, 1), \quad (4, 1). \tag{120}$$

Since $L_x = 4\pi$ and $L_y = \pi$, these correspond to

$$(\kappa, \lambda) = (0.5, 0), \quad (0.5, 2), \quad (1, 0), \quad (1, 2), \quad (2, 2).$$

The first two modes expose the new low streamwise wavenumber made available by the longer box. The last three reproduce the physical wavenumbers used in the earlier 2π-box comparison, whose corresponding indices were $(1, 0)$, $(1, 1)$, and $(2, 1)$. The sparse target-CSD run used the stored target `modal/u_hat` series. The dense temporal-CSD run used selected horizontal modes computed from the dense raw velocity continuation. Table 7 summarizes the two CSD sources.

Table 7: CSD sources used in the completed production selected-mode resolvent workflows.

Run	Source	Record	Δt_s	Segment length	Segments
Sparse target CSD	`modal/u_hat`	$60 \leq t \leq 200$	0.1	512	4
Dense temporal CSD	raw selected FFT modes	$200 \leq t \leq 320$	0.01	2048	10

Both production CSD estimates used a Hann window and selected-record temporal demeaning. The sparse target run used the first four positive CSD bins,

$$\omega = 0.122718,\ 0.245437,\ 0.368155,\ 0.490874, \tag{121}$$

while the dense temporal run used the first six positive dense-CSD bins,

$$\omega = 0.306796,\ 0.613592,\ 0.920388,\ 1.227185,\ 1.533981,\ 1.840777. \tag{122}$$

The two frequency grids are different; the comparison below should therefore be read as a trend and infrastructure check, not as a pointwise same-frequency comparison.

Table 8 gives the sparse target-CSD results. Table 9 gives the dense temporal-CSD results. The gain range is the range of the leading singular value over the stored frequencies. The projection columns report the maximum rank-one fraction F_1 and the maximum cumulative rank-six fraction $F_{\leq 6}$ over the stored frequencies.

Table 8: Sparse target-CSD selected-mode resolvent results.

Mode	(κ, λ)	Leading gain range	$\max F_1$	$\max F_{\leq 6}$
$(1, 0)$	$(0.5, 0)$	0.255031–0.275859	0.130727 at $\omega = 0.245437$	0.439358 at $\omega = 0.490874$
$(1, 1)$	$(0.5, 2)$	0.396685–0.437435	0.230144 at $\omega = 0.122718$	0.482374 at $\omega = 0.245437$
$(2, 0)$	$(1, 0)$	0.145406–0.151780	0.120823 at $\omega = 0.122718$	0.319133 at $\omega = 0.122718$
$(2, 1)$	$(1, 2)$	0.182903–0.192309	0.065530 at $\omega = 0.122718$	0.374398 at $\omega = 0.245437$
$(4, 1)$	$(2, 2)$	0.091810–0.094282	0.088256 at $\omega = 0.245437$	0.282100 at $\omega = 0.245437$

Table 9: Dense temporal-CSD selected-mode resolvent results.

Mode	(κ, λ)	Leading gain range	$\max F_1$	$\max F_{\leq 6}$
$(1, 0)$	$(0.5, 0)$	0.265086–0.384889	0.106173 at $\omega = 0.306796$	0.377876 at $\omega = 0.920388$
$(1, 1)$	$(0.5, 2)$	0.416239–0.665594	0.124887 at $\omega = 0.920388$	0.512384 at $\omega = 0.920388$
$(2, 0)$	$(1, 0)$	0.148534–0.179732	0.094548 at $\omega = 0.920388$	0.349890 at $\omega = 1.227185$
$(2, 1)$	$(1, 2)$	0.187506–0.234880	0.132310 at $\omega = 0.613592$	0.402286 at $\omega = 0.306796$
$(4, 1)$	$(2, 2)$	0.093032–0.104404	0.127032 at $\omega = 0.920388$	0.361420 at $\omega = 0.920388$

In the sparse target-CSD estimate, the new low-streamwise-wavenumber oblique mode $(p, q) = (1, 1)$, corresponding to $(\kappa, \lambda) = (0.5, 2)$, has both the largest maximum rank-one fraction and the largest maximum rank-six fraction. In the dense estimate, the largest rank-one fraction is instead 0.132310 for the physical mode $(\kappa, \lambda) = (1, 2)$ at $\omega = 0.613592$, while the new $(0.5, 2)$ mode retains the largest rank-six fraction, 0.512384, at $\omega = 0.920388$. Rank six captures substantially more energy than rank one: the modewise maxima range from about 28% to 48% in the sparse estimate and from about 35% to 51% in the dense estimate. The two frequency grids do not coincide, so these maxima are not pointwise temporal convergence measurements. A low rank-one fraction does not invalidate the resolvent model; it can reflect forcing statistics not aligned with the leading forcing mode, finite response rank, low Reynolds number, the finite periodic box, selected frequency bins that are not centered on the most energetic CSD bands, and finite-sample CSD estimation.

The numerical diagnostics are near roundoff. The maximum CSD Parseval relative errors were 2.93×10^{-15} for the sparse target-CSD run and 7.20×10^{-15} for the dense temporal-CSD run. Across the resolvent outputs, the largest response constraint residual was 5.25×10^{-15}, the largest forcing constraint residual was 4.88×10^{-15}, the largest response or forcing energy-normalization error was 3.55×10^{-15}, and the projection frequency-match errors were zero for the stored frequencies.

7.9 Representative generated figures

The workflow wrote a complete four-PDF figure set for every selected mode in both the sparse and dense production directories. The local lightweight artifacts are under

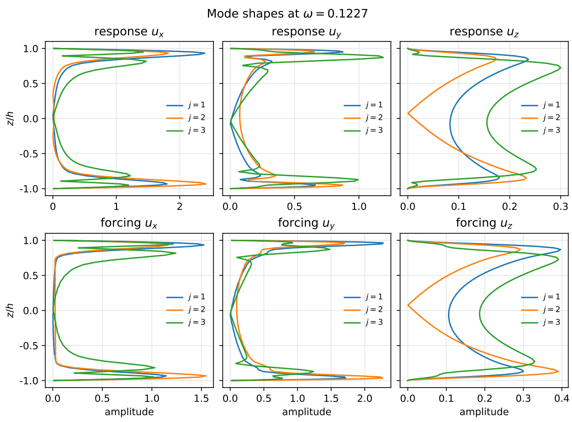
`scripts/mkm_dns_target/production_reports/longbox_Lx4pi_64_128_32/`
`target_csd_selected_modes/`

and

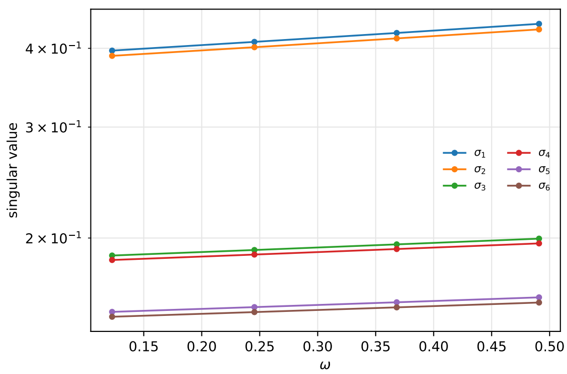
`scripts/mkm_dns_target/production_reports/longbox_Lx4pi_64_128_32/`
`dense_csd_selected_modes/`.

The large HDF5 products remain on the Linux server and were intentionally not copied into the repository.

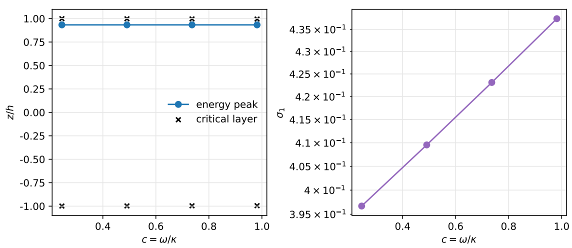
Figures 11 and 12 show the representative new low-streamwise-wavenumber oblique mode $(1,1)$. The mode-shape panels plot $|u_x|$, $|u_y|$, and $|u_z|$ versus z/h for the stored response and forcing modes at the selected frequency index. The gain panels plot the singular values versus ω. The peak-location panels compare the wall-normal location of maximum leading-response energy with stored critical-layer roots. The reconstructed-field panels show real physical velocity slices computed from Eq. (119).



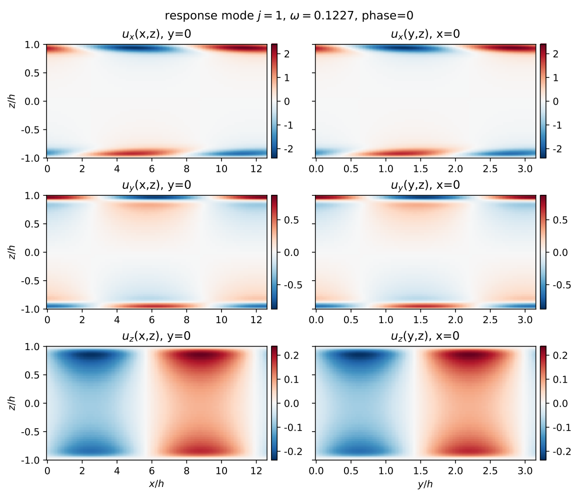
Mode shapes



Singular-value gain curves

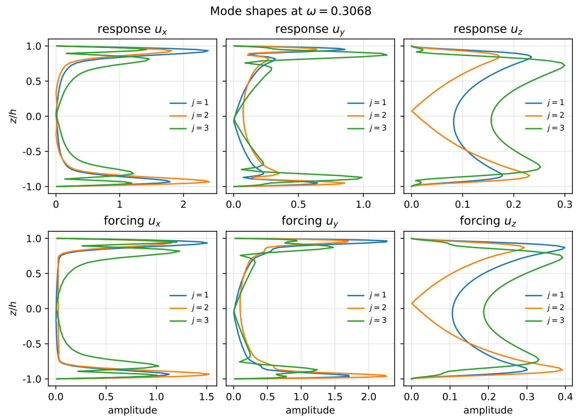


Peak energy and critical layer

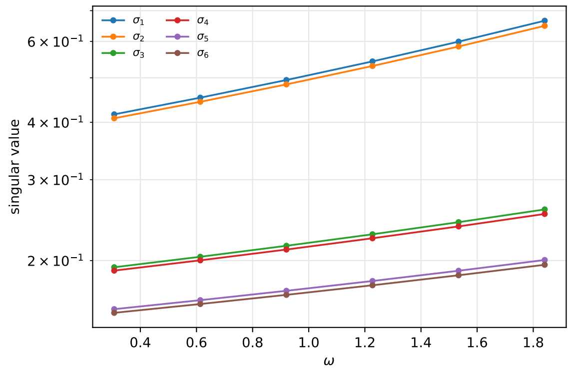


Reconstructed physical fields

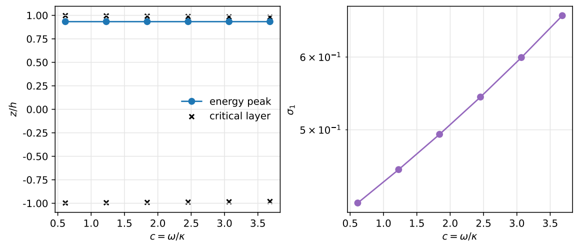
Figure 11: Sparse target-CSD selected-mode resolvent figures for mode $(1,1)$, corresponding to $(\kappa, \lambda) = (0.5, 2)$.



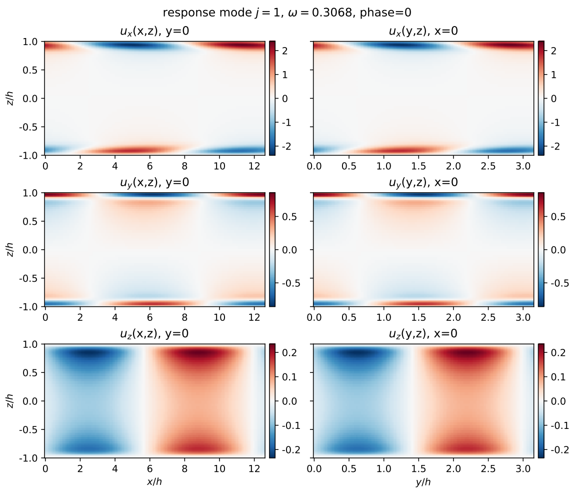
Mode shapes



Singular-value gain curves



Peak energy and critical layer



Reconstructed physical fields

Figure 12: Dense temporal-CSD selected-mode resolvent figures for mode $(1, 1)$, corresponding to $(\kappa, \lambda) = (0.5, 2)$.

The complete local figure set includes the same four PDFs for modes $(1, 0)$, $(1, 1)$, $(2, 0)$, $(2, 1)$, and $(4, 1)$ in both CSD-source directories. The workflow manifest JSON files record the exact HDF5 paths, frequencies, leading singular values, constraint residuals, projection fractions, and figure paths. The generated Markdown reports summarize those manifests and serve as compact audit records for future production reruns.

8 Reproducibility remarks

The production handoff file

`HANDOFF_production_mkm_Lx4pi_64_128_32.md`

records the exact server paths, commands, convergence reports, and audit files. For future runs on this mesh, the validated production setting is 32 MPI ranks. A different MPI decomposition should be re-tested together with the FFT pencil decomposition constraints. If a stricter scalar-energy stationarity criterion is required, the sampling record should be extended beyond $tu_\tau/h = 200$, and the retained-window field and scalar convergence checks should be repeated before regenerating the target covariance.